Supplementary material for "When interval coverage hides the direction of maintenance-trigger errors: a sensor-degradation stress test of remaining-useful-life prediction intervals"

Yi Wang, Lishui University

All tables are generated by `paper_jrr/make_tables_v18.py` (the unchanged v1.8 generator, from released result files) or `paper_jrr/make_tables_jrr.py` (new or re-derived quantities, from released result files and from `results_jrr/derived/`); no value is typed by hand. Rates are percentages of evaluated engines unless stated. Analyses marked *exploratory* were run once after the main protocol was fixed and are not pre-registered. Sections S3–S6 keep the analyses of the first submission, in which perturbation sizes are percentages of each channel's training range (*range scale*); the main text of this version expresses sizes in within-operating-condition standard deviations (SD), and Sections S8–S11 are new.

S1. Protocol audit

Controlled 2×2 retraining. Table 1 trains the heteroscedastic LSTM with the same five seeds under all four combinations of checkpoint selection (test set T or validation engines V) and normaliser scope (whole training file W or fitting engines F), with the fitting engines held fixed. Sel. is the mean over normaliser levels of (T−V), Norm. the mean over selection levels of (W−F), Int. the difference of differences; negative means the choice makes the result look better. Test-set selection improves RMSE by -1.94 to -0.29 cycles and the NASA score by -860 to -8 units; the normaliser scope changes RMSE by -0.04 to 0.07 cycles. The interaction (-0.43 to 0.47 cycles, -700 to -21 score units) is of the same order as the normaliser effect. The only cell in which a leaked choice does not flatter the result is the FD002 score under fit-only normalisation. An earlier uncontrolled version left the fitting set free, so whole-file cells were fitted on more engines, and the normaliser appeared to cost as much as the selection choice; that apparent effect was training-set size.

Table 1. Controlled 2×2 experiment, heteroscedastic LSTM, terminal unit, mean of five seeds.

		T/W	V/W	V/F	T/F	Sel.	Norm.	Int.
FD001	RMSE	13.48	15.31	15.41	13.36	−1.94	+0.01	+0.22
	Score	268.4	558.4	552.2	283.1	−279.5	−4.2	−20.8
FD002	RMSE	16.14	16.20	16.47	15.94	−0.29	−0.04	+0.47
	Score	1342.3	1388.1	1319.5	1349.2	−8.1	+30.8	−75.4
FD004	RMSE	19.35	20.37	20.08	19.49	−0.80	+0.07	−0.43
	Score	2750.5	3960.7	3396.0	2886.3	−859.9	+214.4	−700.4

Audit history. The protocol was adopted after an audit of an earlier, unpublished pipeline found that the checkpoint had been selected on the official test set and the normaliser fitted on data used for checkpoint selection. The audit also found an MC Dropout baseline that omitted the residual variance term (coverage 0.40–0.49) and an ensemble evaluation that gave each member its own noise realisation, which averages the perturbation at $1/\sqrt{M}$ and manufactured apparent robustness. All were corrected; the earlier results are retained in the released repository under `results/superseded/`.

S2. Clean benchmark, mean processing and ensemble controls

Predicted-mean processing by analysis. The analyses of the original pipeline did not post-process the predicted mean identically. Table 2 lists what each analysis does, read from the code; the clean baseline of the main text (Table 1) and Table 3 use the stress-test processing.

Table 2. Predicted-mean processing (clip to $[0, 125]$ or none) by analysis. Heteroscedastic: mean of the heteroscedastic head, also used by CP-norm. Other: fixed-variance, MC Dropout; ensemble mean.

Analysis (scripts)	Heteroscedastic, CP-norm	Other
v1.8 clean benchmark: PICP, MPIW, ECE, per-engine, RMSE (`eval_*_b1`, `eval_*_deep_ensemble`, per-engine coverage)	clipped	clipped; ensemble members clipped before averaging
v1.8 clean interval score of the heteroscedastic head, CP-norm, ensemble (`interval_score.py`)	unclipped	ensemble members unclipped, ensemble mean clipped
v1.8 2×2 retraining, same-split ensemble control (`protocol_2x2_*`, `*samesplit*`)	clipped	clipped
v1.8 sweeps, dose–response, decision tables, drift controls (`sweep_engine`, `maintenance_decision_*`, `drift_controls`)	unclipped	fixed variance and MC Dropout clipped; ensemble members unclipped, mean clipped
Derived analyses of this revision (`jrr_decision_per_engine`, `jrr_coherent_*`)	unclipped	as the sweeps
Clean baseline of this revision (`jrr_clean_same_path.py`)	unclipped (variant B); clipped (variant A) reported for reference	as the sweeps
Clipping sensitivity (`jrr_clip_sensitivity.py`)	both	ensemble members clipped individually (A) or not (B)

Table 3. Clean terminal-window benchmark recomputed through the stress-test code path (mean of five seeds; MC Dropout also over five sampling repeats; the ensemble is a single $M = 5$ estimate). PICP, MPIW (cycles), ECE, interval score IS and RMSE (cycles) use the unclipped heteroscedastic and CP-norm means; the last column is PICP under the v1.8 benchmark processing (all means clipped).

Cell	Mechanism	PICP	MPIW	ECE	IS	RMSE	PICP, v1.8
LSTM FD001	Fixed variance	0.870	47.8	0.062	70.9	15.35	0.870
LSTM FD001	Heteroscedastic	0.864	42.2	0.035	67.7	15.42	0.866
LSTM FD001	MC Dropout	0.891	50.3	0.071	69.5	15.31	0.890
LSTM FD001	Deep ensemble	0.910	45.1	0.031	63.2	14.60	0.910
LSTM FD001	Split-CP (norm)	0.878	45.6	0.038	67.3	15.42	0.880
LSTM FD002	Fixed variance	0.912	60.9	0.059	74.2	17.09	0.912
LSTM FD002	Heteroscedastic	0.913	51.6	0.030	62.4	16.47	0.913
LSTM FD002	MC Dropout	0.927	62.8	0.067	74.4	17.10	0.927
LSTM FD002	Deep ensemble	0.964	54.6	0.058	58.6	15.61	0.964
LSTM FD002	Split-CP (norm)	0.932	53.7	0.036	62.9	16.47	0.932
LSTM FD003	Fixed variance	0.876	46.8	0.104	72.4	14.66	0.876
LSTM FD003	Heteroscedastic	0.858	37.1	0.040	62.8	13.98	0.860
LSTM FD003	MC Dropout	0.890	49.2	0.114	70.7	14.59	0.890
LSTM FD003	Deep ensemble	0.910	40.7	0.078	56.7	13.11	0.910
LSTM FD003	Split-CP (norm)	0.896	46.0	0.082	62.3	13.98	0.898
LSTM FD004	Fixed variance	0.877	60.9	0.051	87.9	19.23	0.877
LSTM FD004	Heteroscedastic	0.882	58.0	0.028	80.7	20.08	0.882
LSTM FD004	MC Dropout	0.889	63.1	0.056	87.5	19.24	0.886
LSTM FD004	Deep ensemble	0.932	61.5	0.042	74.8	19.40	0.932
LSTM FD004	Split-CP (norm)	0.882	56.7	0.046	80.4	20.08	0.882
Transformer FD001	Fixed variance	0.876	45.3	0.079	68.0	14.38	0.876
Transformer FD001	Heteroscedastic	0.880	43.4	0.037	65.9	14.58	0.888
Transformer FD001	MC Dropout	0.916	48.8	0.093	62.5	13.61	0.914
Transformer FD001	Deep ensemble	0.910	45.5	0.065	60.1	13.94	0.910
Transformer FD001	Split-CP (norm)	0.892	43.9	0.063	64.8	14.58	0.896
Transformer FD002	Fixed variance	0.919	58.9	0.080	73.0	16.14	0.919
Transformer FD002	Heteroscedastic	0.892	49.8	0.033	64.8	15.90	0.896
Transformer FD002	MC Dropout	0.921	60.5	0.079	71.8	16.25	0.919
Transformer FD002	Deep ensemble	0.933	52.6	0.046	60.7	15.18	0.933
Transformer FD002	Split-CP (norm)	0.911	52.6	0.032	65.1	15.90	0.915
Transformer FD003	Fixed variance	0.842	42.7	0.090	70.6	14.50	0.842
Transformer FD003	Heteroscedastic	0.870	39.1	0.046	63.5	14.27	0.880
Transformer FD003	MC Dropout	0.882	45.3	0.089	62.6	13.67	0.888
Transformer FD003	Deep ensemble	0.920	43.1	0.083	56.0	13.22	0.930
Transformer FD003	Split-CP (norm)	0.876	44.8	0.110	65.0	14.27	0.884
Transformer FD004	Fixed variance	0.853	55.5	0.074	88.5	18.57	0.853
Transformer FD004	Heteroscedastic	0.857	50.5	0.027	84.1	18.30	0.866
Transformer FD004	MC Dropout	0.882	57.7	0.039	80.9	18.18	0.881
Transformer FD004	Deep ensemble	0.907	55.2	0.057	76.0	17.34	0.907
Transformer FD004	Split-CP (norm)	0.857	51.7	0.119	84.3	18.30	0.866

Table 4. Effect of clipping the heteroscedastic mean to $[0, 125]$ on coverage (stored per-engine outputs; unclipped / clipped), 90% interval, terminal window; noise 1%, bias 5%, drift 5% of range. Trigger rates of the heteroscedastic head and CP-norm are unchanged in all 288 cell–condition–lead-time combinations; the ensemble changes by at most 0.3 points.

Cell	Clean	Noise	Bias	Drift
LSTM FD001	0.864 / 0.866	0.861 / 0.862	0.772 / 0.774	0.644 / 0.644
LSTM FD002	0.913 / 0.913	0.837 / 0.838	0.315 / 0.355	0.048 / 0.111
LSTM FD003	0.858 / 0.860	0.855 / 0.856	0.742 / 0.744	0.214 / 0.214
LSTM FD004	0.882 / 0.882	0.857 / 0.857	0.431 / 0.453	0.311 / 0.321
Transformer FD001	0.880 / 0.888	0.884 / 0.890	0.836 / 0.848	0.642 / 0.644
Transformer FD002	0.892 / 0.896	0.806 / 0.813	0.299 / 0.329	0.025 / 0.111
Transformer FD003	0.870 / 0.880	0.865 / 0.873	0.784 / 0.792	0.096 / 0.122
Transformer FD004	0.857 / 0.866	0.788 / 0.799	0.294 / 0.344	0.053 / 0.127

v1.8 clean benchmark (benchmark processing, unchanged). PICP, MPIW, ECE and per-engine compliance are from the clipped benchmark; the interval score of the heteroscedastic head, CP-norm and ensemble is the unclipped-path value from the interval-score file, and that of the fixed-variance and MC Dropout heads comes from the fair-calibration files, so the IS column mixes processings (compare Table 3).

Table 5. v1.8 clean-input benchmark under fair calibration, v1.8 processing, terminal unit except per-engine (trajectory unit), mean of five seeds; the ensemble is a single $M = 5$ estimate.

		LSTM					Transformer				
	Mechanism	PICP	MPIW	ECE	Per-eng.	IS	PICP	MPIW	ECE	Per-eng.	IS
FD001	Fixed variance	0.870	47.82	0.062	0.55	70.8	0.876	45.26	0.079	0.61	67.9
	Heteroscedastic	0.866	42.20	0.035	0.69	67.7	0.888	43.40	0.041	0.77	65.9
	MC Dropout	0.890	50.31	0.072	0.62	69.2	0.914	48.85	0.093	0.72	62.9
	Deep ensemble	0.910	45.05	0.032	0.81	63.1	0.910	45.40	0.065	0.81	60.1
	Split-CP (norm)	0.880	45.60	0.038	0.75	67.2	0.896	43.85	0.056	0.75	64.7
FD002	Fixed variance	0.912	60.92	0.059	0.60	74.1	0.919	58.91	0.080	0.63	72.9
	Heteroscedastic	0.913	51.63	0.030	0.75	62.4	0.896	49.79	0.032	0.73	64.7
	MC Dropout	0.927	62.82	0.067	0.64	74.3	0.919	60.52	0.079	0.63	71.8
	Deep ensemble	0.964	54.63	0.058	0.81	58.6	0.933	52.52	0.046	0.80	60.6
	Split-CP (norm)	0.932	53.72	0.037	0.77	62.9	0.915	52.60	0.028	0.76	65.1
FD003	Fixed variance	0.876	46.80	0.104	0.70	72.4	0.842	42.71	0.090	0.67	70.5
	Heteroscedastic	0.860	37.08	0.042	0.72	62.7	0.880	39.14	0.048	0.76	63.4
	MC Dropout	0.890	49.15	0.114	0.72	71.0	0.888	45.29	0.088	0.72	62.4
	Deep ensemble	0.910	40.63	0.076	0.76	56.7	0.930	42.86	0.082	0.84	56.0
	Split-CP (norm)	0.898	46.05	0.073	0.78	62.2	0.884	44.76	0.085	0.83	65.0
FD004	Fixed variance	0.877	60.93	0.051	0.68	87.8	0.853	55.53	0.074	0.68	88.4
	Heteroscedastic	0.882	58.01	0.027	0.76	80.6	0.866	50.48	0.029	0.75	84.0
	MC Dropout	0.886	63.07	0.056	0.70	87.4	0.881	57.66	0.038	0.74	81.0
	Deep ensemble	0.932	61.47	0.042	0.81	74.8	0.907	55.06	0.055	0.82	75.9
	Split-CP (norm)	0.882	56.72	0.046	0.73	80.3	0.866	51.74	0.094	0.76	84.2

Table 6. Ensemble gain with the partition held fixed. Single models: five initialisations on the seed-42 partition; same-partition ensemble: those five combined; cross-partition ensemble: the five seeds of the main benchmark, each on its own partition.

			IS			PICP	
Backbone	Data	5 part., single	1 part., single	1 part., ens.	5 part., ens.	1 part., ens.	5 part., ens.
LSTM	FD001	67.7	60.2 ± 2.2	58.0	63.1	0.890	0.910
LSTM	FD002	62.4	63.5 ± 3.9	58.7	58.6	0.917	0.964
LSTM	FD003	62.7	61.2 ± 5.3	57.2	56.7	0.920	0.910
LSTM	FD004	80.6	81.7 ± 3.6	77.7	74.8	0.928	0.932
Transformer	FD001	65.9	61.8 ± 3.2	59.5	60.1	0.910	0.910
Transformer	FD002	64.7	67.1 ± 6.0	62.2	60.6	0.925	0.933
Transformer	FD003	63.4	64.3 ± 6.0	55.3	56.0	0.920	0.930
Transformer	FD004	84.0	83.4 ± 4.5	75.7	75.9	0.899	0.907

The same-partition ensemble beats the mean of its own members by 2.1–9.0 IS points in all eight combinations, and the two ensembles differ by at most 5.1 points with no consistent direction. On FD001 five single models on one partition average 60.2 IS against 67.7 for five single models on five partitions, a difference as large as the ensemble margin; five seeds on five partitions are therefore a coarse instrument. Three disjoint five-member ensembles from a fifteen-seed pool give IS of 56.7/55.1/60.3, 60.4/58.7/56.8, 52.2/54.2/55.8, 75.2/74.8/77.4 on FD001–FD004; the ratio of fifteen- to five-seed PICP standard deviation is 1.36, 0.70, 0.80 on FD001, FD002 and FD004; over $M \in \{2, 3, 5, 10\}$ the FD004 PICP rises from 0.909 to 0.924 and IS falls from 77.6 to 75.0.

Fair calibration. Table 7 regenerates the calibration-versus-training residual comparison from released result files. The v1.8 manuscript typed this table by hand; its ratio column agrees with the result files, but its coverage columns did not agree with

the main benchmark (differences up to 0.016), and the training-residual coverage is available from the released results only for the LSTM fixed-variance head on FD001, FD002 and FD004.

Table 7. Ratio of calibration-set to training-residual scale, and PICP under each estimator ("–": not in the released result files).

	Ratio	PICP fixed, train	PICP fixed, calibration	PICP MC Dropout, calibration
LSTM FD001	1.241	0.790	0.870	0.890
LSTM FD002	1.125	0.874	0.912	0.927
LSTM FD003	1.364	–	0.876	0.890
LSTM FD004	1.103	0.850	0.877	0.886
Transformer FD001	1.079	–	0.876	0.914
Transformer FD002	1.084	–	0.919	0.919
Transformer FD003	1.238	–	0.842	0.888
Transformer FD004	1.168	–	0.853	0.881

S3. Coverage against out-of-range share (range scale)

Perturbation levels are percentages of the training range for noise, bias and drift and percentages of the reading for gain. Matched-excursion interpolation (heteroscedastic head, linear in f_{oob} between measured levels; "–": the target lies outside the measured range of that curve). The underlying points are in `results_jrr/derived/foob_coverage_points.csv`.

Table 8. Coverage at matched f_{oob} of 5% and 10% (excursion levels, not perturbation amplitudes; amplitudes are percentages of range except gain, which is a percentage of the reading).

Backbone	Data	f_{oob}	Noise	Bias	Gain	Drift
LSTM	FD001	5%	–	–	–	–
LSTM	FD001	10%	–	–	–	–
LSTM	FD002	5%	0.88	0.88	–	0.64
LSTM	FD002	10%	0.68	0.78	0.28	0.14
LSTM	FD003	5%	–	–	–	–
LSTM	FD003	10%	–	–	–	–
LSTM	FD004	5%	0.86	0.85	0.81	0.82
LSTM	FD004	10%	0.72	0.64	0.46	–
Transformer	FD001	5%	–	–	–	–
Transformer	FD001	10%	–	–	–	–
Transformer	FD002	5%	0.85	0.85	0.56	0.58
Transformer	FD002	10%	0.62	0.71	0.21	0.09
Transformer	FD003	5%	–	–	–	–
Transformer	FD003	10%	–	–	–	–
Transformer	FD004	5%	0.79	0.81	0.36	0.57
Transformer	FD004	10%	0.55	0.52	0.24	–

Table 9. Excursion f_{oob} at which the heteroscedastic head loses ten points of coverage (v1.8 definition), with left-censored cells marked LC (coverage already lost ten points at the lowest level; the f_{oob} at that level in parentheses). n.r.: not reached within 5% of range; $< 10^{-4}$: coverage falls while the excursion stays below 10^{-4}.

Backbone	Degradation	FD001	FD002	FD003	FD004
LSTM	Bias	n.r.	9.47%	n.r.	8.24%
	Drift	0.00%	2.53%	$< 10^{-4}$	5.40%
	Gain	LC (14%)	6.35%	LC (14%)	5.18%
Transformer	Bias	n.r.	8.30%	0.03%	6.70%
	Drift	0.00%	2.22%	$< 10^{-4}$	2.94%
	Gain	LC (14%)	LC (4%)	LC (14%)	4.15%

S4. Decision stress test (range scale)

Table 10. Both mechanisms' families at $L = 20$ for drift and bias, six-condition sub-datasets; engine-bootstrap 95% intervals (%).

Cell	Condition	Mechanism	Premature	Unrecognised
LSTM FD002	Drift	Fixed variance	70 [64, 75]	0.0 [0.0, 0.0]
LSTM FD002	Drift	Heteroscedastic	69 [64, 75]	0.0 [0.0, 0.0]
LSTM FD002	Drift	MC Dropout	70 [64, 75]	0.0 [0.0, 0.0]
LSTM FD002	Drift	Deep ensemble	70 [64, 75]	0.0 [0.0, 0.0]
LSTM FD002	Drift	Split-CP (norm)	69 [63, 75]	0.0 [0.0, 0.0]
LSTM FD002	Bias	Fixed variance	14 [13, 15]	7.2 [5.4, 9.2]
LSTM FD002	Bias	Heteroscedastic	17 [15, 18]	7.8 [5.8, 10.0]
LSTM FD002	Bias	MC Dropout	14 [13, 15]	7.1 [5.2, 9.0]
LSTM FD002	Bias	Deep ensemble	19 [17, 21]	4.5 [3.0, 6.1]
LSTM FD002	Bias	Split-CP (norm)	17 [16, 18]	7.1 [5.2, 9.0]
LSTM FD004	Drift	Fixed variance	63 [58, 69]	0.0 [0.0, 0.0]
LSTM FD004	Drift	Heteroscedastic	59 [53, 64]	0.0 [0.0, 0.0]
LSTM FD004	Drift	MC Dropout	63 [58, 69]	0.0 [0.0, 0.0]
LSTM FD004	Drift	Deep ensemble	65 [59, 71]	0.0 [0.0, 0.0]
LSTM FD004	Drift	Split-CP (norm)	64 [58, 70]	0.0 [0.0, 0.0]
LSTM FD004	Bias	Fixed variance	34 [31, 37]	2.6 [1.7, 3.7]
LSTM FD004	Bias	Heteroscedastic	30 [27, 33]	2.5 [1.7, 3.5]
LSTM FD004	Bias	MC Dropout	34 [31, 38]	2.5 [1.6, 3.6]
LSTM FD004	Bias	Deep ensemble	35 [31, 38]	1.4 [0.6, 2.1]
LSTM FD004	Bias	Split-CP (norm)	32 [29, 35]	2.2 [1.5, 3.1]
Transformer FD002	Drift	Fixed variance	70 [64, 75]	0.0 [0.0, 0.0]
Transformer FD002	Drift	Heteroscedastic	69 [64, 74]	0.0 [0.0, 0.0]
Transformer FD002	Drift	MC Dropout	70 [64, 75]	0.0 [0.0, 0.0]
Transformer FD002	Drift	Deep ensemble	70 [64, 75]	0.0 [0.0, 0.0]
Transformer FD002	Drift	Split-CP (norm)	69 [64, 75]	0.0 [0.0, 0.0]
Transformer FD002	Bias	Fixed variance	14 [13, 15]	7.6 [5.7, 9.8]
Transformer FD002	Bias	Heteroscedastic	14 [13, 15]	10.2 [7.7, 13.1]
Transformer FD002	Bias	MC Dropout	14 [13, 15]	7.9 [6.0, 10.2]
Transformer FD002	Bias	Deep ensemble	14 [13, 16]	5.8 [4.0, 8.0]
Transformer FD002	Bias	Split-CP (norm)	14 [13, 15]	9.2 [6.9, 11.8]
Transformer FD004	Drift	Fixed variance	66 [60, 72]	0.0 [0.0, 0.0]
Transformer FD004	Drift	Heteroscedastic	52 [47, 58]	0.0 [0.0, 0.0]
Transformer FD004	Drift	MC Dropout	64 [59, 70]	0.0 [0.0, 0.0]
Transformer FD004	Drift	Deep ensemble	57 [51, 64]	0.0 [0.0, 0.0]
Transformer FD004	Drift	Split-CP (norm)	56 [51, 62]	0.0 [0.0, 0.0]
Transformer FD004	Bias	Fixed variance	36 [32, 39]	1.6 [1.0, 2.2]
Transformer FD004	Bias	Heteroscedastic	32 [29, 35]	1.8 [1.2, 2.4]
Transformer FD004	Bias	MC Dropout	35 [32, 38]	1.9 [1.2, 2.6]
Transformer FD004	Bias	Deep ensemble	35 [32, 38]	0.6 [0.2, 1.0]
Transformer FD004	Bias	Split-CP (norm)	34 [31, 37]	1.5 [0.9, 2.1]

Table 11. Heteroscedastic head, all backbones, sub-datasets, conditions and lead times; premature and unrecognised rates (%) with engine-bootstrap 95% intervals.

Backbone	Data	Condition	L	Premature	Unrecognised
LSTM	FD001	Clean	10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
LSTM	FD001	Clean	20	0.0 [0.0, 0.0]	0.2 [0.0, 0.6]
LSTM	FD001	Clean	30	0.8 [0.0, 2.0]	0.0 [0.0, 0.0]
LSTM	FD001	Noise 1%	10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
LSTM	FD001	Noise 1%	20	0.0 [0.0, 0.0]	0.2 [0.0, 0.6]
LSTM	FD001	Noise 1%	30	0.7 [0.0, 1.9]	0.1 [0.0, 0.2]
LSTM	FD001	Drift 5%	10	1.8 [0.0, 4.6]	0.0 [0.0, 0.0]
LSTM	FD001	Drift 5%	20	2.6 [0.4, 5.4]	0.0 [0.0, 0.0]
LSTM	FD001	Drift 5%	30	6.8 [2.8, 11.8]	0.0 [0.0, 0.0]
LSTM	FD001	Bias 5%	10	0.3 [0.0, 0.8]	0.0 [0.0, 0.0]
LSTM	FD001	Bias 5%	20	0.1 [0.0, 0.2]	0.5 [0.1, 1.1]
LSTM	FD001	Bias 5%	30	1.4 [0.5, 2.8]	0.3 [0.0, 0.7]
LSTM	FD002	Clean	10	0.1 [0.0, 0.2]	0.0 [0.0, 0.0]
LSTM	FD002	Clean	20	3.7 [2.1, 5.5]	0.0 [0.0, 0.0]
LSTM	FD002	Clean	30	7.4 [4.9, 10.3]	0.0 [0.0, 0.0]
LSTM	FD002	Noise 1%	10	0.7 [0.3, 1.1]	0.3 [0.1, 0.6]
LSTM	FD002	Noise 1%	20	5.0 [3.3, 6.9]	0.2 [0.0, 0.5]
LSTM	FD002	Noise 1%	30	9.4 [7.1, 12.0]	0.0 [0.0, 0.1]
LSTM	FD002	Drift 5%	10	71.5 [66.2, 76.6]	0.0 [0.0, 0.0]
LSTM	FD002	Drift 5%	20	69.3 [63.6, 74.9]	0.0 [0.0, 0.0]
LSTM	FD002	Drift 5%	30	64.3 [58.5, 70.4]	0.0 [0.0, 0.0]
LSTM	FD002	Bias 5%	10	17.1 [15.8, 18.5]	5.5 [3.7, 7.7]
LSTM	FD002	Bias 5%	20	17.0 [15.5, 18.4]	7.8 [5.8, 10.0]
LSTM	FD002	Bias 5%	30	16.1 [14.6, 17.7]	5.9 [4.3, 7.6]
LSTM	FD003	Clean	10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
LSTM	FD003	Clean	20	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
LSTM	FD003	Clean	30	1.4 [0.2, 3.0]	0.0 [0.0, 0.0]
LSTM	FD003	Noise 1%	10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]

Backbone	Data	Condition	L	Premature	Unrecognised
LSTM	FD003	Noise 1%	20	0.1 [0.0, 0.2]	0.0 [0.0, 0.0]
LSTM	FD003	Noise 1%	30	1.6 [0.2, 3.4]	0.0 [0.0, 0.0]
LSTM	FD003	Drift 5%	10	2.2 [0.4, 4.8]	0.0 [0.0, 0.0]
LSTM	FD003	Drift 5%	20	13.8 [8.0, 19.8]	0.0 [0.0, 0.0]
LSTM	FD003	Drift 5%	30	25.0 [18.0, 32.8]	0.0 [0.0, 0.0]
LSTM	FD003	Bias 5%	10	0.2 [0.0, 0.7]	0.2 [0.0, 0.4]
LSTM	FD003	Bias 5%	20	0.6 [0.2, 1.0]	0.1 [0.0, 0.2]
LSTM	FD003	Bias 5%	30	3.8 [1.7, 6.3]	0.1 [0.0, 0.3]
LSTM	FD004	Clean	10	3.3 [1.7, 5.1]	0.3 [0.0, 0.8]
LSTM	FD004	Clean	20	3.9 [2.0, 6.0]	0.1 [0.0, 0.3]
LSTM	FD004	Clean	30	6.8 [4.2, 9.8]	0.3 [0.0, 0.8]
LSTM	FD004	Noise 1%	10	4.4 [2.6, 6.2]	0.6 [0.0, 1.4]
LSTM	FD004	Noise 1%	20	5.5 [3.6, 7.7]	0.3 [0.0, 0.7]
LSTM	FD004	Noise 1%	30	8.6 [6.2, 11.1]	0.3 [0.0, 0.6]
LSTM	FD004	Drift 5%	10	46.6 [41.9, 51.4]	0.0 [0.0, 0.0]
LSTM	FD004	Drift 5%	20	59.0 [53.2, 64.5]	0.0 [0.0, 0.0]
LSTM	FD004	Drift 5%	30	61.9 [55.9, 67.5]	0.0 [0.0, 0.0]
LSTM	FD004	Bias 5%	10	28.9 [26.3, 31.4]	1.6 [0.9, 2.5]
LSTM	FD004	Bias 5%	20	30.2 [27.2, 33.0]	2.5 [1.7, 3.5]
LSTM	FD004	Bias 5%	30	32.3 [28.9, 35.3]	3.0 [2.1, 4.0]
Transformer	FD001	Clean	10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Transformer	FD001	Clean	20	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Transformer	FD001	Clean	30	0.8 [0.0, 2.4]	0.0 [0.0, 0.0]
Transformer	FD001	Noise 1%	10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Transformer	FD001	Noise 1%	20	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Transformer	FD001	Noise 1%	30	0.8 [0.0, 2.4]	0.0 [0.0, 0.0]
Transformer	FD001	Drift 5%	10	1.6 [0.0, 4.0]	0.0 [0.0, 0.0]
Transformer	FD001	Drift 5%	20	2.4 [0.4, 5.0]	0.0 [0.0, 0.0]
Transformer	FD001	Drift 5%	30	6.2 [2.8, 10.4]	0.0 [0.0, 0.0]
Transformer	FD001	Bias 5%	10	0.2 [0.0, 0.7]	0.0 [0.0, 0.0]
Transformer	FD001	Bias 5%	20	0.0 [0.0, 0.0]	0.2 [0.0, 0.4]
Transformer	FD001	Bias 5%	30	1.0 [0.2, 2.3]	0.0 [0.0, 0.1]
Transformer	FD002	Clean	10	0.0 [0.0, 0.0]	0.3 [0.0, 0.9]
Transformer	FD002	Clean	20	1.5 [0.5, 2.8]	0.1 [0.0, 0.2]
Transformer	FD002	Clean	30	6.0 [3.8, 8.6]	0.0 [0.0, 0.0]
Transformer	FD002	Noise 1%	10	0.5 [0.1, 1.1]	0.3 [0.0, 0.7]
Transformer	FD002	Noise 1%	20	3.1 [1.9, 4.3]	0.2 [0.0, 0.5]
Transformer	FD002	Noise 1%	30	8.1 [6.0, 10.6]	0.0 [0.0, 0.0]
Transformer	FD002	Drift 5%	10	69.1 [63.5, 74.0]	0.0 [0.0, 0.0]
Transformer	FD002	Drift 5%	20	69.2 [63.6, 74.5]	0.0 [0.0, 0.0]
Transformer	FD002	Drift 5%	30	64.3 [58.3, 70.0]	0.0 [0.0, 0.0]
Transformer	FD002	Bias 5%	10	13.9 [12.9, 14.9]	6.5 [4.2, 9.4]
Transformer	FD002	Bias 5%	20	13.8 [12.7, 14.9]	10.2 [7.7, 13.1]
Transformer	FD002	Bias 5%	30	13.3 [12.1, 14.6]	8.2 [6.2, 10.5]
Transformer	FD003	Clean	10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Transformer	FD003	Clean	20	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Transformer	FD003	Clean	30	2.2 [0.6, 4.2]	0.0 [0.0, 0.0]
Transformer	FD003	Noise 1%	10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Transformer	FD003	Noise 1%	20	0.0 [0.0, 0.1]	0.0 [0.0, 0.0]
Transformer	FD003	Noise 1%	30	2.2 [0.7, 4.0]	0.0 [0.0, 0.0]
Transformer	FD003	Drift 5%	10	4.6 [1.6, 8.2]	0.0 [0.0, 0.0]
Transformer	FD003	Drift 5%	20	17.4 [10.8, 24.4]	0.0 [0.0, 0.0]
Transformer	FD003	Drift 5%	30	30.6 [22.8, 38.8]	0.0 [0.0, 0.0]
Transformer	FD003	Bias 5%	10	0.2 [0.0, 0.6]	0.0 [0.0, 0.1]
Transformer	FD003	Bias 5%	20	0.5 [0.2, 1.0]	0.0 [0.0, 0.0]
Transformer	FD003	Bias 5%	30	3.8 [1.7, 6.2]	0.0 [0.0, 0.0]
Transformer	FD004	Clean	10	3.0 [1.6, 4.7]	0.0 [0.0, 0.0]
Transformer	FD004	Clean	20	2.9 [1.4, 4.6]	0.0 [0.0, 0.0]
Transformer	FD004	Clean	30	4.7 [2.6, 7.1]	0.0 [0.0, 0.0]
Transformer	FD004	Noise 1%	10	3.8 [2.2, 5.6]	0.1 [0.0, 0.2]
Transformer	FD004	Noise 1%	20	3.9 [2.4, 5.6]	0.0 [0.0, 0.1]
Transformer	FD004	Noise 1%	30	7.1 [4.9, 9.5]	0.1 [0.0, 0.1]
Transformer	FD004	Drift 5%	10	40.3 [34.7, 46.2]	0.0 [0.0, 0.0]
Transformer	FD004	Drift 5%	20	52.5 [46.8, 58.4]	0.0 [0.0, 0.0]
Transformer	FD004	Drift 5%	30	60.6 [54.8, 67.0]	0.0 [0.0, 0.0]
Transformer	FD004	Bias 5%	10	32.0 [29.4, 34.5]	0.9 [0.4, 1.4]
Transformer	FD004	Bias 5%	20	31.9 [28.9, 34.8]	1.8 [1.2, 2.4]
Transformer	FD004	Bias 5%	30	33.3 [29.9, 36.8]	2.1 [1.4, 2.8]

Table 12. Rank stability across cost ratios 5/20/100 by condition: groups (backbone $\times$ sub-dataset $\times$ lead time), groups with a changed full ranking, groups with a changed top place, and the median engine-bootstrap probability that the top mechanism at ratio 5 differs from that at ratio 100 among groups with a changed top place.

Condition	Groups	Ranking changes	Top changes	Median probability
Clean	24	7	6	0.45
Noise 1%	24	13	11	0.85
Drift 5%	24	0	0	–
Bias 5%	24	16	3	0.86

Full cost and rank table (v1.8 generator). Cost $= c_{\text{late}} r_{\text{late}} + r_{\text{early}}$ at ratios 5/20/100 and rank for every mechanism. In this table unrecognised (overall / conditional) and premature rates are fractions between 0 and 1, not percentages; cost is in the same units; RUL@t is the true RUL at premature trigger under the 125-cycle clip. The table is generated by the unchanged v1.8 generator. Rates and engine-bootstrap intervals of all five mechanisms, eight cells, four conditions and three lead times are enumerated in `results_jrr/derived/decision_engine_bootstrap_all.csv`.

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
LSTM	FD001	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Clean	10	Fixed var.	0.0000 / 0.000	0.0180	37.4	0.018 / 0.018 / 0.018	4/4/4
LSTM	FD001	Clean	10	MC Drop.	0.0000 / 0.000	0.0180	37.4	0.018 / 0.018 / 0.018	5/5/5
LSTM	FD001	Clean	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD001	Clean	10	CP-norm	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	3/3/3
LSTM	FD001	Clean	20	Heterosc.	0.0020 / 0.011	0.0000	–	0.010 / 0.040 / 0.200	3/5/5
LSTM	FD001	Clean	20	Fixed var.	0.0000 / 0.000	0.0220	54.6	0.022 / 0.022 / 0.022	4/3/3
LSTM	FD001	Clean	20	MC Drop.	0.0000 / 0.000	0.0220	54.6	0.022 / 0.022 / 0.022	5/4/4
LSTM	FD001	Clean	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Clean	20	CP-norm	0.0000 / 0.000	0.0020	46.0	0.002 / 0.002 / 0.002	2/2/2
LSTM	FD001	Clean	30	Heterosc.	0.0000 / 0.000	0.0080	60.5	0.008 / 0.008 / 0.008	2/2/2
LSTM	FD001	Clean	30	Fixed var.	0.0000 / 0.000	0.0300	68.3	0.030 / 0.030 / 0.030	4/4/4
LSTM	FD001	Clean	30	MC Drop.	0.0000 / 0.000	0.0300	68.3	0.030 / 0.030 / 0.030	5/5/5
LSTM	FD001	Clean	30	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Clean	30	CP-norm	0.0000 / 0.000	0.0220	69.7	0.022 / 0.022 / 0.022	3/3/3
LSTM	FD001	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0172	36.7	0.017 / 0.017 / 0.017	4/4/4
LSTM	FD001	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0180	37.4	0.018 / 0.018 / 0.018	5/5/5
LSTM	FD001	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD001	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0004	33.0	0.000 / 0.000 / 0.000	3/3/3
LSTM	FD001	Noise 1%	20	Heterosc.	0.0020 / 0.011	0.0000	–	0.010 / 0.040 / 0.200	3/5/5
LSTM	FD001	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0208	54.3	0.021 / 0.021 / 0.021	4/3/2
LSTM	FD001	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0232	55.8	0.023 / 0.023 / 0.023	5/4/3
LSTM	FD001	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Noise 1%	20	CP-norm	0.0004 / 0.002	0.0020	46.0	0.004 / 0.010 / 0.042	2/2/4
LSTM	FD001	Noise 1%	30	Heterosc.	0.0008 / 0.003	0.0072	62.2	0.011 / 0.023 / 0.087	2/3/5
LSTM	FD001	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0296	68.1	0.030 / 0.030 / 0.030	4/4/3
LSTM	FD001	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0300	68.3	0.030 / 0.030 / 0.030	5/5/4
LSTM	FD001	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0020	68.0	0.002 / 0.002 / 0.002	1/1/1
LSTM	FD001	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0188	65.8	0.019 / 0.019 / 0.019	3/2/2
LSTM	FD001	Bias 5%	10	Heterosc.	0.0000 / 0.000	0.0028	37.0	0.003 / 0.003 / 0.003	1/1/1
LSTM	FD001	Bias 5%	10	Fixed var.	0.0000 / 0.000	0.0188	38.4	0.019 / 0.019 / 0.019	4/4/4
LSTM	FD001	Bias 5%	10	MC Drop.	0.0000 / 0.000	0.0192	39.0	0.019 / 0.019 / 0.019	5/5/5
LSTM	FD001	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0040	37.0	0.004 / 0.004 / 0.004	2/2/2
LSTM	FD001	Bias 5%	10	CP-norm	0.0000 / 0.000	0.0056	39.5	0.006 / 0.006 / 0.006	3/3/3
LSTM	FD001	Bias 5%	20	Heterosc.	0.0052 / 0.027	0.0008	57.0	0.027 / 0.105 / 0.521	3/5/5
LSTM	FD001	Bias 5%	20	Fixed var.	0.0016 / 0.008	0.0264	59.2	0.034 / 0.058 / 0.186	5/3/3
LSTM	FD001	Bias 5%	20	MC Drop.	0.0012 / 0.006	0.0272	58.9	0.033 / 0.051 / 0.147	4/2/2
LSTM	FD001	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD001	Bias 5%	20	CP-norm	0.0028 / 0.015	0.0120	75.1	0.026 / 0.068 / 0.292	2/4/4
LSTM	FD001	Bias 5%	30	Heterosc.	0.0032 / 0.013	0.0144	68.4	0.030 / 0.078 / 0.334	2/5/5
LSTM	FD001	Bias 5%	30	Fixed var.	0.0012 / 0.005	0.0316	69.1	0.038 / 0.056 / 0.152	3/2/2
LSTM	FD001	Bias 5%	30	MC Drop.	0.0012 / 0.005	0.0336	69.3	0.040 / 0.058 / 0.154	4/3/3
LSTM	FD001	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0220	68.1	0.022 / 0.022 / 0.022	1/1/1
LSTM	FD001	Bias 5%	30	CP-norm	0.0016 / 0.006	0.0384	77.3	0.046 / 0.070 / 0.198	5/4/4
LSTM	FD001	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0180	35.2	0.018 / 0.018 / 0.018	1/1/1
LSTM	FD001	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.0720	48.4	0.072 / 0.072 / 0.072	4/4/4
LSTM	FD001	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.0740	49.4	0.074 / 0.074 / 0.074	5/5/5

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
LSTM	FD001	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0200	35.0	0.020 / 0.020 / 0.020	2/2/2
LSTM	FD001	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0280	38.3	0.028 / 0.028 / 0.028	3/3/3
LSTM	FD001	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.0260	62.0	0.026 / 0.026 / 0.026	1/1/1
LSTM	FD001	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.1180	61.5	0.118 / 0.118 / 0.118	4/4/4
LSTM	FD001	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.1200	61.3	0.120 / 0.120 / 0.120	5/5/5
LSTM	FD001	Drift 5%	20	Ensemble	0.0000 / 0.000	0.0300	54.3	0.030 / 0.030 / 0.030	2/2/2
LSTM	FD001	Drift 5%	20	CP-norm	0.0000 / 0.000	0.0580	66.8	0.058 / 0.058 / 0.058	3/3/3
LSTM	FD001	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.0680	68.3	0.068 / 0.068 / 0.068	1/1/1
LSTM	FD001	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.1360	77.8	0.136 / 0.136 / 0.136	3/3/3
LSTM	FD001	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.1400	77.6	0.140 / 0.140 / 0.140	4/4/4
LSTM	FD001	Drift 5%	30	Ensemble	0.0000 / 0.000	0.1600	78.1	0.160 / 0.160 / 0.160	5/5/5
LSTM	FD001	Drift 5%	30	CP-norm	0.0000 / 0.000	0.1340	80.9	0.134 / 0.134 / 0.134	2/2/2
LSTM	FD002	Clean	10	Heterosc.	0.0000 / 0.000	0.0008	36.0	0.001 / 0.001 / 0.001	1/1/1
LSTM	FD002	Clean	10	Fixed var.	0.0000 / 0.000	0.0617	42.4	0.062 / 0.062 / 0.062	4/4/4
LSTM	FD002	Clean	10	MC Drop.	0.0000 / 0.000	0.0625	42.4	0.062 / 0.062 / 0.062	5/5/5
LSTM	FD002	Clean	10	Ensemble	0.0000 / 0.000	0.0040	42.0	0.004 / 0.004 / 0.004	2/2/2
LSTM	FD002	Clean	10	CP-norm	0.0000 / 0.000	0.0095	48.8	0.009 / 0.009 / 0.009	3/3/3
LSTM	FD002	Clean	20	Heterosc.	0.0000 / 0.000	0.0372	49.4	0.037 / 0.037 / 0.037	1/1/1
LSTM	FD002	Clean	20	Fixed var.	0.0000 / 0.000	0.0783	52.6	0.078 / 0.078 / 0.078	4/4/4
LSTM	FD002	Clean	20	MC Drop.	0.0000 / 0.000	0.0814	53.0	0.081 / 0.081 / 0.081	5/5/5
LSTM	FD002	Clean	20	Ensemble	0.0000 / 0.000	0.0514	48.2	0.051 / 0.051 / 0.051	2/2/2
LSTM	FD002	Clean	20	CP-norm	0.0000 / 0.000	0.0522	54.3	0.052 / 0.052 / 0.052	3/3/3
LSTM	FD002	Clean	30	Heterosc.	0.0000 / 0.000	0.0743	68.0	0.074 / 0.074 / 0.074	3/3/3
LSTM	FD002	Clean	30	Fixed var.	0.0000 / 0.000	0.0719	64.0	0.072 / 0.072 / 0.072	1/1/1
LSTM	FD002	Clean	30	MC Drop.	0.0000 / 0.000	0.0719	64.2	0.072 / 0.072 / 0.072	2/2/2
LSTM	FD002	Clean	30	Ensemble	0.0000 / 0.000	0.0870	67.1	0.087 / 0.087 / 0.087	5/5/5
LSTM	FD002	Clean	30	CP-norm	0.0000 / 0.000	0.0838	69.8	0.084 / 0.084 / 0.084	4/4/4
LSTM	FD002	Noise 1%	10	Heterosc.	0.0032 / 0.029	0.0068	44.3	0.023 / 0.070 / 0.323	1/2/4
LSTM	FD002	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0757	46.3	0.076 / 0.076 / 0.076	4/3/1
LSTM	FD002	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0778	46.8	0.078 / 0.078 / 0.078	5/5/2
LSTM	FD002	Noise 1%	10	Ensemble	0.0032 / 0.029	0.0126	39.8	0.028 / 0.076 / 0.329	2/4/5
LSTM	FD002	Noise 1%	10	CP-norm	0.0027 / 0.024	0.0150	50.7	0.028 / 0.069 / 0.284	3/1/3
LSTM	FD002	Noise 1%	20	Heterosc.	0.0022 / 0.010	0.0496	56.1	0.061 / 0.094 / 0.271	1/2/5
LSTM	FD002	Noise 1%	20	Fixed var.	0.0003 / 0.001	0.0877	56.8	0.089 / 0.094 / 0.119	4/3/1
LSTM	FD002	Noise 1%	20	MC Drop.	0.0003 / 0.001	0.0889	57.0	0.090 / 0.095 / 0.120	5/4/2
LSTM	FD002	Noise 1%	20	Ensemble	0.0008 / 0.004	0.0704	57.3	0.074 / 0.086 / 0.149	3/1/3
LSTM	FD002	Noise 1%	20	CP-norm	0.0016 / 0.007	0.0653	61.8	0.073 / 0.097 / 0.223	2/5/4
LSTM	FD002	Noise 1%	30	Heterosc.	0.0003 / 0.001	0.0938	77.8	0.095 / 0.100 / 0.125	3/3/5
LSTM	FD002	Noise 1%	30	Fixed var.	0.0002 / 0.001	0.0822	71.4	0.083 / 0.085 / 0.098	1/2/2
LSTM	FD002	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0841	71.8	0.084 / 0.084 / 0.084	2/1/1
LSTM	FD002	Noise 1%	30	Ensemble	0.0000 / 0.000	0.1075	77.2	0.108 / 0.108 / 0.108	5/5/3
LSTM	FD002	Noise 1%	30	CP-norm	0.0002 / 0.001	0.1018	78.5	0.103 / 0.105 / 0.118	4/4/4
LSTM	FD002	Bias 5%	10	Heterosc.	0.0553 / 0.500	0.1711	89.2	0.448 / 1.278 / 5.705	5/5/5
LSTM	FD002	Bias 5%	10	Fixed var.	0.0370 / 0.334	0.1521	90.5	0.337 / 0.892 / 3.852	2/2/2
LSTM	FD002	Bias 5%	10	MC Drop.	0.0360 / 0.326	0.1521	90.5	0.332 / 0.873 / 3.757	1/1/1
LSTM	FD002	Bias 5%	10	Ensemble	0.0395 / 0.357	0.1684	86.6	0.366 / 0.959 / 4.121	3/3/3
LSTM	FD002	Bias 5%	10	CP-norm	0.0536 / 0.484	0.1700	88.6	0.438 / 1.242 / 5.530	4/4/4
LSTM	FD002	Bias 5%	20	Heterosc.	0.0783 / 0.367	0.1698	94.3	0.561 / 1.735 / 7.996	5/5/5
LSTM	FD002	Bias 5%	20	Fixed var.	0.0721 / 0.338	0.1410	95.0	0.502 / 1.583 / 7.351	3/3/4
LSTM	FD002	Bias 5%	20	MC Drop.	0.0705 / 0.330	0.1412	94.9	0.494 / 1.551 / 7.193	2/2/2
LSTM	FD002	Bias 5%	20	Ensemble	0.0451 / 0.211	0.1905	92.9	0.416 / 1.092 / 4.696	1/1/1
LSTM	FD002	Bias 5%	20	CP-norm	0.0707 / 0.331	0.1698	93.9	0.523 / 1.583 / 7.237	4/4/3
LSTM	FD002	Bias 5%	30	Heterosc.	0.0590 / 0.241	0.1609	99.1	0.456 / 1.340 / 6.058	3/3/3
LSTM	FD002	Bias 5%	30	Fixed var.	0.0751 / 0.306	0.1300	99.2	0.505 / 1.632 / 7.640	5/5/5
LSTM	FD002	Bias 5%	30	MC Drop.	0.0729 / 0.297	0.1303	99.1	0.495 / 1.588 / 7.419	4/4/4

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
LSTM	FD002	Bias 5%	30	Ensemble	0.0348 / 0.142	0.1897	98.1	0.364 / 0.885 / 3.668	1/1/1
LSTM	FD002	Bias 5%	30	CP-norm	0.0582 / 0.237	0.1606	98.8	0.452 / 1.324 / 5.979	2/2/2
LSTM	FD002	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.7154	89.4	0.715 / 0.715 / 0.715	2/2/2
LSTM	FD002	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.7518	90.8	0.752 / 0.752 / 0.752	4/4/4
LSTM	FD002	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.7518	90.8	0.752 / 0.752 / 0.752	5/5/5
LSTM	FD002	Drift 5%	10	Ensemble	0.0000 / 0.000	0.7431	90.5	0.743 / 0.743 / 0.743	3/3/3
LSTM	FD002	Drift 5%	10	CP-norm	0.0000 / 0.000	0.7115	89.3	0.711 / 0.711 / 0.711	1/1/1
LSTM	FD002	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.6933	95.1	0.693 / 0.693 / 0.693	2/2/2
LSTM	FD002	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.6996	95.3	0.700 / 0.700 / 0.700	4/4/4
LSTM	FD002	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.6996	95.3	0.700 / 0.700 / 0.700	5/5/5
LSTM	FD002	Drift 5%	20	Ensemble	0.0000 / 0.000	0.6957	95.2	0.696 / 0.696 / 0.696	3/3/3
LSTM	FD002	Drift 5%	20	CP-norm	0.0000 / 0.000	0.6917	95.0	0.692 / 0.692 / 0.692	1/1/1
LSTM	FD002	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6435	99.6	0.643 / 0.643 / 0.643	1/1/1
LSTM	FD002	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	3/3/3
LSTM	FD002	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	4/4/4
LSTM	FD002	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	5/5/5
LSTM	FD002	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6435	99.6	0.643 / 0.643 / 0.643	2/2/2
LSTM	FD003	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Clean	10	Fixed var.	0.0000 / 0.000	0.0160	48.1	0.016 / 0.016 / 0.016	5/5/5
LSTM	FD003	Clean	10	MC Drop.	0.0000 / 0.000	0.0140	45.0	0.014 / 0.014 / 0.014	4/4/4
LSTM	FD003	Clean	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD003	Clean	10	CP-norm	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	3/3/3
LSTM	FD003	Clean	20	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Clean	20	Fixed var.	0.0000 / 0.000	0.0380	54.5	0.038 / 0.038 / 0.038	4/4/4
LSTM	FD003	Clean	20	MC Drop.	0.0000 / 0.000	0.0420	54.0	0.042 / 0.042 / 0.042	5/5/5
LSTM	FD003	Clean	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD003	Clean	20	CP-norm	0.0000 / 0.000	0.0320	63.6	0.032 / 0.032 / 0.032	3/3/3
LSTM	FD003	Clean	30	Heterosc.	0.0000 / 0.000	0.0140	60.9	0.014 / 0.014 / 0.014	1/1/1
LSTM	FD003	Clean	30	Fixed var.	0.0000 / 0.000	0.0440	59.5	0.044 / 0.044 / 0.044	3/3/3
LSTM	FD003	Clean	30	MC Drop.	0.0000 / 0.000	0.0500	59.1	0.050 / 0.050 / 0.050	4/4/4
LSTM	FD003	Clean	30	Ensemble	0.0000 / 0.000	0.0300	59.3	0.030 / 0.030 / 0.030	2/2/2
LSTM	FD003	Clean	30	CP-norm	0.0000 / 0.000	0.0680	64.4	0.068 / 0.068 / 0.068	5/5/5
LSTM	FD003	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0148	46.4	0.015 / 0.015 / 0.015	4/4/4
LSTM	FD003	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0148	46.4	0.015 / 0.015 / 0.015	5/5/5
LSTM	FD003	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
LSTM	FD003	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0012	68.0	0.001 / 0.001 / 0.001	3/3/3
LSTM	FD003	Noise 1%	20	Heterosc.	0.0000 / 0.000	0.0008	54.0	0.001 / 0.001 / 0.001	2/2/2
LSTM	FD003	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0340	54.7	0.034 / 0.034 / 0.034	4/4/4
LSTM	FD003	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0352	54.3	0.035 / 0.035 / 0.035	5/5/5
LSTM	FD003	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
LSTM	FD003	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0292	63.5	0.029 / 0.029 / 0.029	3/3/3
LSTM	FD003	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0160	60.8	0.016 / 0.016 / 0.016	1/1/1
LSTM	FD003	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0496	59.3	0.050 / 0.050 / 0.050	3/3/3
LSTM	FD003	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0536	59.0	0.054 / 0.054 / 0.054	4/4/4
LSTM	FD003	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0340	58.7	0.034 / 0.034 / 0.034	2/2/2
LSTM	FD003	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0740	65.1	0.074 / 0.074 / 0.074	5/5/5
LSTM	FD003	Bias 5%	10	Heterosc.	0.0016 / 0.020	0.0024	36.7	0.010 / 0.034 / 0.162	3/3/5
LSTM	FD003	Bias 5%	10	Fixed var.	0.0008 / 0.010	0.0300	46.4	0.034 / 0.046 / 0.110	4/4/3
LSTM	FD003	Bias 5%	10	MC Drop.	0.0008 / 0.010	0.0328	47.5	0.037 / 0.049 / 0.113	5/5/4
LSTM	FD003	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0020	34.0	0.002 / 0.002 / 0.002	1/1/1
LSTM	FD003	Bias 5%	10	CP-norm	0.0004 / 0.005	0.0048	49.2	0.007 / 0.013 / 0.045	2/2/2
LSTM	FD003	Bias 5%	20	Heterosc.	0.0008 / 0.006	0.0060	55.2	0.010 / 0.022 / 0.086	2/2/3
LSTM	FD003	Bias 5%	20	Fixed var.	0.0008 / 0.006	0.0572	54.7	0.061 / 0.073 / 0.137	4/4/4
LSTM	FD003	Bias 5%	20	MC Drop.	0.0008 / 0.006	0.0592	54.8	0.063 / 0.075 / 0.139	5/5/5

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
LSTM	FD003	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0060	57.3	0.006 / 0.006 / 0.006	1/1/1
LSTM	FD003	Bias 5%	20	CP-norm	0.0000 / 0.000	0.0472	63.2	0.047 / 0.047 / 0.047	3/3/2
LSTM	FD003	Bias 5%	30	Heterosc.	0.0012 / 0.006	0.0384	59.5	0.044 / 0.062 / 0.158	1/2/4
LSTM	FD003	Bias 5%	30	Fixed var.	0.0008 / 0.004	0.0696	62.5	0.074 / 0.086 / 0.150	3/4/3
LSTM	FD003	Bias 5%	30	MC Drop.	0.0004 / 0.002	0.0728	62.4	0.075 / 0.081 / 0.113	4/3/2
LSTM	FD003	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0620	60.7	0.062 / 0.062 / 0.062	2/1/1
LSTM	FD003	Bias 5%	30	CP-norm	0.0020 / 0.010	0.1004	67.9	0.110 / 0.140 / 0.300	5/5/5
LSTM	FD003	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0220	47.4	0.022 / 0.022 / 0.022	2/2/2
LSTM	FD003	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.2360	56.2	0.236 / 0.236 / 0.236	4/4/4
LSTM	FD003	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.2400	56.3	0.240 / 0.240 / 0.240	5/5/5
LSTM	FD003	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0200	52.0	0.020 / 0.020 / 0.020	1/1/1
LSTM	FD003	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0440	46.1	0.044 / 0.044 / 0.044	3/3/3
LSTM	FD003	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.1380	55.1	0.138 / 0.138 / 0.138	1/1/1
LSTM	FD003	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.3540	68.7	0.354 / 0.354 / 0.354	4/4/4
LSTM	FD003	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.3600	69.1	0.360 / 0.360 / 0.360	5/5/5
LSTM	FD003	Drift 5%	20	Ensemble	0.0000 / 0.000	0.1600	55.9	0.160 / 0.160 / 0.160	2/2/2
LSTM	FD003	Drift 5%	20	CP-norm	0.0000 / 0.000	0.1880	60.7	0.188 / 0.188 / 0.188	3/3/3
LSTM	FD003	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.2500	71.3	0.250 / 0.250 / 0.250	1/1/1
LSTM	FD003	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.4220	82.6	0.422 / 0.422 / 0.422	4/4/4
LSTM	FD003	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.4280	82.7	0.428 / 0.428 / 0.428	5/5/5
LSTM	FD003	Drift 5%	30	Ensemble	0.0000 / 0.000	0.2600	70.5	0.260 / 0.260 / 0.260	2/2/2
LSTM	FD003	Drift 5%	30	CP-norm	0.0000 / 0.000	0.3540	80.9	0.354 / 0.354 / 0.354	3/3/3
LSTM	FD004	Clean	10	Heterosc.	0.0034 / 0.044	0.0329	42.6	0.050 / 0.100 / 0.370	1/5/5
LSTM	FD004	Clean	10	Fixed var.	0.0000 / 0.000	0.0726	42.7	0.073 / 0.073 / 0.073	4/3/3
LSTM	FD004	Clean	10	MC Drop.	0.0000 / 0.000	0.0734	42.8	0.073 / 0.073 / 0.073	5/4/4
LSTM	FD004	Clean	10	Ensemble	0.0000 / 0.000	0.0506	43.8	0.051 / 0.051 / 0.051	2/1/1
LSTM	FD004	Clean	10	CP-norm	0.0000 / 0.000	0.0650	47.9	0.065 / 0.065 / 0.065	3/2/2
LSTM	FD004	Clean	20	Heterosc.	0.0008 / 0.005	0.0388	60.0	0.043 / 0.056 / 0.123	1/2/3
LSTM	FD004	Clean	20	Fixed var.	0.0017 / 0.010	0.0363	57.1	0.045 / 0.070 / 0.205	2/4/4
LSTM	FD004	Clean	20	MC Drop.	0.0017 / 0.010	0.0388	56.9	0.047 / 0.073 / 0.208	4/5/5
LSTM	FD004	Clean	20	Ensemble	0.0000 / 0.000	0.0464	60.5	0.046 / 0.046 / 0.046	3/1/1
LSTM	FD004	Clean	20	CP-norm	0.0000 / 0.000	0.0675	69.7	0.068 / 0.068 / 0.068	5/3/2
LSTM	FD004	Clean	30	Heterosc.	0.0025 / 0.011	0.0684	77.6	0.081 / 0.119 / 0.322	3/1/2
LSTM	FD004	Clean	30	Fixed var.	0.0042 / 0.019	0.0371	72.6	0.058 / 0.122 / 0.459	1/2/3
LSTM	FD004	Clean	30	MC Drop.	0.0042 / 0.019	0.0380	72.4	0.059 / 0.122 / 0.460	2/3/4
LSTM	FD004	Clean	30	Ensemble	0.0042 / 0.019	0.0675	75.0	0.089 / 0.152 / 0.489	4/5/5
LSTM	FD004	Clean	30	CP-norm	0.0008 / 0.004	0.1055	80.7	0.110 / 0.122 / 0.190	5/4/1
LSTM	FD004	Noise 1%	10	Heterosc.	0.0056 / 0.073	0.0441	47.9	0.072 / 0.155 / 0.601	2/5/5
LSTM	FD004	Noise 1%	10	Fixed var.	0.0010 / 0.013	0.0734	45.0	0.078 / 0.094 / 0.175	3/2/2
LSTM	FD004	Noise 1%	10	MC Drop.	0.0008 / 0.011	0.0751	45.3	0.079 / 0.092 / 0.159	4/1/1
LSTM	FD004	Noise 1%	10	Ensemble	0.0025 / 0.033	0.0523	46.3	0.065 / 0.103 / 0.305	1/3/3
LSTM	FD004	Noise 1%	10	CP-norm	0.0029 / 0.038	0.0759	53.6	0.090 / 0.133 / 0.363	5/4/4
LSTM	FD004	Noise 1%	20	Heterosc.	0.0025 / 0.015	0.0550	67.2	0.068 / 0.106 / 0.308	3/4/5
LSTM	FD004	Noise 1%	20	Fixed var.	0.0025 / 0.015	0.0523	62.2	0.065 / 0.103 / 0.305	2/3/4
LSTM	FD004	Noise 1%	20	MC Drop.	0.0022 / 0.013	0.0538	62.1	0.065 / 0.098 / 0.273	1/2/3
LSTM	FD004	Noise 1%	20	Ensemble	0.0017 / 0.010	0.0616	66.0	0.070 / 0.095 / 0.230	4/1/2
LSTM	FD004	Noise 1%	20	CP-norm	0.0014 / 0.008	0.0906	73.8	0.097 / 0.118 / 0.226	5/5/1
LSTM	FD004	Noise 1%	30	Heterosc.	0.0027 / 0.012	0.0857	81.4	0.099 / 0.140 / 0.356	3/1/3
LSTM	FD004	Noise 1%	30	Fixed var.	0.0054 / 0.024	0.0557	77.7	0.083 / 0.164 / 0.596	1/5/5
LSTM	FD004	Noise 1%	30	MC Drop.	0.0052 / 0.023	0.0582	78.2	0.084 / 0.163 / 0.581	2/4/4
LSTM	FD004	Noise 1%	30	Ensemble	0.0025 / 0.011	0.0928	78.9	0.105 / 0.143 / 0.346	4/2/2
LSTM	FD004	Noise 1%	30	CP-norm	0.0014 / 0.006	0.1200	83.9	0.127 / 0.147 / 0.255	5/3/1
LSTM	FD004	Bias 5%	10	Heterosc.	0.0164 / 0.216	0.2889	83.7	0.371 / 0.616 / 1.926	1/3/5
LSTM	FD004	Bias 5%	10	Fixed var.	0.0140 / 0.184	0.3507	86.2	0.421 / 0.631 / 1.752	4/5/3
LSTM	FD004	Bias 5%	10	MC Drop.	0.0138 / 0.182	0.3529	86.2	0.422 / 0.630 / 1.737	5/4/2

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
LSTM	FD004	Bias 5%	10	Ensemble	0.0110 / 0.144	0.3367	86.7	0.392 / 0.556 / 1.434	3/1/1
LSTM	FD004	Bias 5%	10	CP-norm	0.0145 / 0.191	0.3151	84.4	0.388 / 0.605 / 1.767	2/2/4
LSTM	FD004	Bias 5%	20	Heterosc.	0.0255 / 0.151	0.3016	94.5	0.429 / 0.811 / 2.850	2/3/3
LSTM	FD004	Bias 5%	20	Fixed var.	0.0257 / 0.152	0.3411	95.1	0.469 / 0.854 / 2.906	4/5/5
LSTM	FD004	Bias 5%	20	MC Drop.	0.0255 / 0.151	0.3431	95.1	0.471 / 0.853 / 2.892	5/4/4
LSTM	FD004	Bias 5%	20	Ensemble	0.0135 / 0.080	0.3468	96.2	0.414 / 0.617 / 1.697	1/1/1
LSTM	FD004	Bias 5%	20	CP-norm	0.0224 / 0.133	0.3219	94.7	0.434 / 0.771 / 2.567	3/2/2
LSTM	FD004	Bias 5%	30	Heterosc.	0.0300 / 0.132	0.3229	99.3	0.473 / 0.924 / 3.327	2/3/3
LSTM	FD004	Bias 5%	30	Fixed var.	0.0322 / 0.141	0.3389	99.4	0.500 / 0.984 / 3.563	5/5/5
LSTM	FD004	Bias 5%	30	MC Drop.	0.0305 / 0.134	0.3414	99.5	0.494 / 0.952 / 3.396	4/4/4
LSTM	FD004	Bias 5%	30	Ensemble	0.0186 / 0.081	0.3519	100.2	0.445 / 0.723 / 2.208	1/1/1
LSTM	FD004	Bias 5%	30	CP-norm	0.0278 / 0.122	0.3399	99.3	0.479 / 0.897 / 3.125	3/2/2
LSTM	FD004	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.4658	81.8	0.466 / 0.466 / 0.466	1/1/1
LSTM	FD004	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.6329	87.3	0.633 / 0.633 / 0.633	3/3/3
LSTM	FD004	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.6397	87.6	0.640 / 0.640 / 0.640	5/5/5
LSTM	FD004	Drift 5%	10	Ensemble	0.0000 / 0.000	0.6371	88.3	0.637 / 0.637 / 0.637	4/4/4
LSTM	FD004	Drift 5%	10	CP-norm	0.0000 / 0.000	0.5882	86.9	0.588 / 0.588 / 0.588	2/2/2
LSTM	FD004	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.5899	96.0	0.590 / 0.590 / 0.590	1/1/1
LSTM	FD004	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.6346	97.2	0.635 / 0.635 / 0.635	2/2/2
LSTM	FD004	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.6346	97.2	0.635 / 0.635 / 0.635	3/3/3
LSTM	FD004	Drift 5%	20	Ensemble	0.0000 / 0.000	0.6540	97.6	0.654 / 0.654 / 0.654	5/5/5
LSTM	FD004	Drift 5%	20	CP-norm	0.0000 / 0.000	0.6422	97.5	0.642 / 0.642 / 0.642	4/4/4
LSTM	FD004	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6194	101.2	0.619 / 0.619 / 0.619	1/1/1
LSTM	FD004	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.6253	101.3	0.625 / 0.625 / 0.625	2/2/2
LSTM	FD004	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6278	101.3	0.628 / 0.628 / 0.628	3/3/3
LSTM	FD004	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6287	101.3	0.629 / 0.629 / 0.629	4/4/4
LSTM	FD004	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6304	101.5	0.630 / 0.630 / 0.630	5/5/5
Transformer	FD001	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Clean	10	Fixed var.	0.0000 / 0.000	0.0220	36.0	0.022 / 0.022 / 0.022	5/5/5
Transformer	FD001	Clean	10	MC Drop.	0.0000 / 0.000	0.0200	36.3	0.020 / 0.020 / 0.020	4/4/4
Transformer	FD001	Clean	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Clean	10	CP-norm	0.0000 / 0.000	0.0040	35.0	0.004 / 0.004 / 0.004	3/3/3
Transformer	FD001	Clean	20	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Clean	20	Fixed var.	0.0000 / 0.000	0.0200	49.6	0.020 / 0.020 / 0.020	4/4/4
Transformer	FD001	Clean	20	MC Drop.	0.0000 / 0.000	0.0260	49.3	0.026 / 0.026 / 0.026	5/5/5
Transformer	FD001	Clean	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Clean	20	CP-norm	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	3/3/3
Transformer	FD001	Clean	30	Heterosc.	0.0000 / 0.000	0.0080	53.0	0.008 / 0.008 / 0.008	1/1/1
Transformer	FD001	Clean	30	Fixed var.	0.0000 / 0.000	0.0220	67.0	0.022 / 0.022 / 0.022	3/3/3
Transformer	FD001	Clean	30	MC Drop.	0.0000 / 0.000	0.0240	67.1	0.024 / 0.024 / 0.024	4/4/4
Transformer	FD001	Clean	30	Ensemble	0.0000 / 0.000	0.0100	53.0	0.010 / 0.010 / 0.010	2/2/2
Transformer	FD001	Clean	30	CP-norm	0.0000 / 0.000	0.0380	68.8	0.038 / 0.038 / 0.038	5/5/5
Transformer	FD001	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0212	36.1	0.021 / 0.021 / 0.021	5/5/5
Transformer	FD001	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0208	36.6	0.021 / 0.021 / 0.021	4/4/4
Transformer	FD001	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0040	35.4	0.004 / 0.004 / 0.004	3/3/3
Transformer	FD001	Noise 1%	20	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0208	49.6	0.021 / 0.021 / 0.021	4/4/4
Transformer	FD001	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0260	50.9	0.026 / 0.026 / 0.026	5/5/5
Transformer	FD001	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD001	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	3/3/3
Transformer	FD001	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0084	53.7	0.008 / 0.008 / 0.008	2/2/2
Transformer	FD001	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0224	67.6	0.022 / 0.022 / 0.022	3/3/3
Transformer	FD001	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0244	67.9	0.024 / 0.024 / 0.024	4/4/4

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
Transformer	FD001	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0060	53.0	0.006 / 0.006 / 0.006	1/1/1
Transformer	FD001	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0376	69.0	0.038 / 0.038 / 0.038	5/5/5
Transformer	FD001	Bias 5%	10	Heterosc.	0.0000 / 0.000	0.0024	36.3	0.002 / 0.002 / 0.002	2/2/2
Transformer	FD001	Bias 5%	10	Fixed var.	0.0000 / 0.000	0.0188	36.4	0.019 / 0.019 / 0.019	4/4/4
Transformer	FD001	Bias 5%	10	MC Drop.	0.0000 / 0.000	0.0204	38.1	0.020 / 0.020 / 0.020	5/5/5
Transformer	FD001	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0020	37.0	0.002 / 0.002 / 0.002	1/1/1
Transformer	FD001	Bias 5%	10	CP-norm	0.0000 / 0.000	0.0048	36.0	0.005 / 0.005 / 0.005	3/3/3
Transformer	FD001	Bias 5%	20	Heterosc.	0.0016 / 0.008	0.0000	–	0.008 / 0.032 / 0.160	3/5/5
Transformer	FD001	Bias 5%	20	Fixed var.	0.0000 / 0.000	0.0208	53.2	0.021 / 0.021 / 0.021	4/3/3
Transformer	FD001	Bias 5%	20	MC Drop.	0.0000 / 0.000	0.0236	53.2	0.024 / 0.024 / 0.024	5/4/4
Transformer	FD001	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD001	Bias 5%	20	CP-norm	0.0000 / 0.000	0.0008	64.5	0.001 / 0.001 / 0.001	2/2/2
Transformer	FD001	Bias 5%	30	Heterosc.	0.0004 / 0.002	0.0104	60.8	0.012 / 0.018 / 0.050	2/2/5
Transformer	FD001	Bias 5%	30	Fixed var.	0.0000 / 0.000	0.0236	64.2	0.024 / 0.024 / 0.024	3/3/2
Transformer	FD001	Bias 5%	30	MC Drop.	0.0000 / 0.000	0.0260	64.7	0.026 / 0.026 / 0.026	4/4/3
Transformer	FD001	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0100	59.0	0.010 / 0.010 / 0.010	1/1/1
Transformer	FD001	Bias 5%	30	CP-norm	0.0000 / 0.000	0.0316	69.5	0.032 / 0.032 / 0.032	5/5/4
Transformer	FD001	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0160	35.9	0.016 / 0.016 / 0.016	1/1/1
Transformer	FD001	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.0680	47.0	0.068 / 0.068 / 0.068	4/4/4
Transformer	FD001	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.0700	47.0	0.070 / 0.070 / 0.070	5/5/5
Transformer	FD001	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0200	35.0	0.020 / 0.020 / 0.020	3/3/3
Transformer	FD001	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0160	35.9	0.016 / 0.016 / 0.016	2/2/2
Transformer	FD001	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.0240	54.5	0.024 / 0.024 / 0.024	2/2/2
Transformer	FD001	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.1020	58.5	0.102 / 0.102 / 0.102	4/4/4
Transformer	FD001	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.1040	59.1	0.104 / 0.104 / 0.104	5/5/5
Transformer	FD001	Drift 5%	20	Ensemble	0.0000 / 0.000	0.0200	57.0	0.020 / 0.020 / 0.020	1/1/1
Transformer	FD001	Drift 5%	20	CP-norm	0.0000 / 0.000	0.0360	64.1	0.036 / 0.036 / 0.036	3/3/3
Transformer	FD001	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.0620	72.4	0.062 / 0.062 / 0.062	1/1/1
Transformer	FD001	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.1320	75.3	0.132 / 0.132 / 0.132	4/4/4
Transformer	FD001	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.1420	75.8	0.142 / 0.142 / 0.142	5/5/5
Transformer	FD001	Drift 5%	30	Ensemble	0.0000 / 0.000	0.1300	72.8	0.130 / 0.130 / 0.130	2/2/2
Transformer	FD001	Drift 5%	30	CP-norm	0.0000 / 0.000	0.1300	83.1	0.130 / 0.130 / 0.130	3/3/3
Transformer	FD002	Clean	10	Heterosc.	0.0032 / 0.029	0.0000	–	0.016 / 0.063 / 0.316	1/3/3
Transformer	FD002	Clean	10	Fixed var.	0.0000 / 0.000	0.0427	40.6	0.043 / 0.043 / 0.043	5/2/2
Transformer	FD002	Clean	10	MC Drop.	0.0000 / 0.000	0.0411	40.3	0.041 / 0.041 / 0.041	4/1/1
Transformer	FD002	Clean	10	Ensemble	0.0040 / 0.036	0.0000	–	0.020 / 0.079 / 0.395	2/4/4
Transformer	FD002	Clean	10	CP-norm	0.0040 / 0.036	0.0055	44.0	0.025 / 0.085 / 0.401	3/5/5
Transformer	FD002	Clean	20	Heterosc.	0.0008 / 0.004	0.0150	46.7	0.019 / 0.031 / 0.094	1/2/4
Transformer	FD002	Clean	20	Fixed var.	0.0000 / 0.000	0.0625	51.9	0.062 / 0.062 / 0.062	4/4/2
Transformer	FD002	Clean	20	MC Drop.	0.0000 / 0.000	0.0648	51.9	0.065 / 0.065 / 0.065	5/5/3
Transformer	FD002	Clean	20	Ensemble	0.0000 / 0.000	0.0198	44.2	0.020 / 0.020 / 0.020	2/1/1
Transformer	FD002	Clean	20	CP-norm	0.0008 / 0.004	0.0316	52.9	0.036 / 0.047 / 0.111	3/3/5
Transformer	FD002	Clean	30	Heterosc.	0.0000 / 0.000	0.0601	69.9	0.060 / 0.060 / 0.060	1/1/1
Transformer	FD002	Clean	30	Fixed var.	0.0000 / 0.000	0.0632	63.7	0.063 / 0.063 / 0.063	2/2/2
Transformer	FD002	Clean	30	MC Drop.	0.0000 / 0.000	0.0680	64.1	0.068 / 0.068 / 0.068	4/4/4
Transformer	FD002	Clean	30	Ensemble	0.0000 / 0.000	0.0711	65.3	0.071 / 0.071 / 0.071	5/5/5
Transformer	FD002	Clean	30	CP-norm	0.0000 / 0.000	0.0664	70.1	0.066 / 0.066 / 0.066	3/3/3
Transformer	FD002	Noise 1%	10	Heterosc.	0.0028 / 0.026	0.0051	38.0	0.019 / 0.062 / 0.290	2/3/4
Transformer	FD002	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0667	47.5	0.067 / 0.067 / 0.067	5/4/2
Transformer	FD002	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0610	46.3	0.061 / 0.061 / 0.061	4/2/1
Transformer	FD002	Noise 1%	10	Ensemble	0.0008 / 0.007	0.0079	42.0	0.012 / 0.024 / 0.087	1/1/3
Transformer	FD002	Noise 1%	10	CP-norm	0.0030 / 0.027	0.0115	43.9	0.027 / 0.072 / 0.312	3/5/5
Transformer	FD002	Noise 1%	20	Heterosc.	0.0021 / 0.010	0.0308	56.2	0.041 / 0.072 / 0.236	1/2/4
Transformer	FD002	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0817	60.2	0.082 / 0.082 / 0.082	5/4/2
Transformer	FD002	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0816	59.8	0.082 / 0.082 / 0.082	4/3/1

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
Transformer	FD002	Noise 1%	20	Ensemble	0.0008 / 0.004	0.0395	55.8	0.043 / 0.055 / 0.119	2/1/3
Transformer	FD002	Noise 1%	20	CP-norm	0.0021 / 0.010	0.0474	62.2	0.058 / 0.089 / 0.253	3/5/5
Transformer	FD002	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0813	77.3	0.081 / 0.081 / 0.081	1/1/1
Transformer	FD002	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0889	73.2	0.089 / 0.089 / 0.089	2/2/2
Transformer	FD002	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0920	73.5	0.092 / 0.092 / 0.092	4/4/4
Transformer	FD002	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0909	75.2	0.091 / 0.091 / 0.091	3/3/3
Transformer	FD002	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0963	78.7	0.096 / 0.096 / 0.096	5/5/5
Transformer	FD002	Bias 5%	10	Heterosc.	0.0655 / 0.591	0.1393	88.7	0.467 / 1.448 / 6.685	4/5/5
Transformer	FD002	Bias 5%	10	Fixed var.	0.0411 / 0.371	0.1507	90.1	0.356 / 0.973 / 4.261	1/1/1
Transformer	FD002	Bias 5%	10	MC Drop.	0.0443 / 0.400	0.1499	90.2	0.371 / 1.035 / 4.577	2/2/2
Transformer	FD002	Bias 5%	10	Ensemble	0.0530 / 0.479	0.1502	90.8	0.415 / 1.209 / 5.447	3/3/3
Transformer	FD002	Bias 5%	10	CP-norm	0.0643 / 0.581	0.1458	87.8	0.468 / 1.433 / 6.581	5/4/4
Transformer	FD002	Bias 5%	20	Heterosc.	0.1021 / 0.479	0.1380	94.3	0.649 / 2.181 / 10.351	5/5/5
Transformer	FD002	Bias 5%	20	Fixed var.	0.0759 / 0.356	0.1432	94.0	0.523 / 1.661 / 7.732	2/2/2
Transformer	FD002	Bias 5%	20	MC Drop.	0.0794 / 0.372	0.1421	94.4	0.539 / 1.729 / 8.079	3/3/3
Transformer	FD002	Bias 5%	20	Ensemble	0.0585 / 0.274	0.1447	93.9	0.437 / 1.315 / 5.994	1/1/1
Transformer	FD002	Bias 5%	20	CP-norm	0.0919 / 0.430	0.1428	93.4	0.602 / 1.980 / 9.329	4/4/4
Transformer	FD002	Bias 5%	30	Heterosc.	0.0821 / 0.335	0.1334	98.4	0.544 / 1.775 / 8.339	5/5/5
Transformer	FD002	Bias 5%	30	Fixed var.	0.0776 / 0.317	0.1345	97.9	0.523 / 1.687 / 7.897	4/4/4
Transformer	FD002	Bias 5%	30	MC Drop.	0.0773 / 0.315	0.1344	97.9	0.521 / 1.681 / 7.866	3/3/3
Transformer	FD002	Bias 5%	30	Ensemble	0.0530 / 0.216	0.1407	96.8	0.406 / 1.200 / 5.437	1/1/1
Transformer	FD002	Bias 5%	30	CP-norm	0.0729 / 0.297	0.1363	98.1	0.501 / 1.594 / 7.425	2/2/2
Transformer	FD002	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.6909	88.4	0.691 / 0.691 / 0.691	1/1/1
Transformer	FD002	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.7502	90.8	0.750 / 0.750 / 0.750	5/5/5
Transformer	FD002	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.7494	90.7	0.749 / 0.749 / 0.749	4/4/4
Transformer	FD002	Drift 5%	10	Ensemble	0.0000 / 0.000	0.7154	89.4	0.715 / 0.715 / 0.715	3/3/3
Transformer	FD002	Drift 5%	10	CP-norm	0.0000 / 0.000	0.7043	89.0	0.704 / 0.704 / 0.704	2/2/2
Transformer	FD002	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.6917	95.0	0.692 / 0.692 / 0.692	1/1/1
Transformer	FD002	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.6988	95.3	0.699 / 0.699 / 0.699	3/3/3
Transformer	FD002	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.6988	95.3	0.699 / 0.699 / 0.699	4/4/4
Transformer	FD002	Drift 5%	20	Ensemble	0.0000 / 0.000	0.6996	95.3	0.700 / 0.700 / 0.700	5/5/5
Transformer	FD002	Drift 5%	20	CP-norm	0.0000 / 0.000	0.6925	95.0	0.692 / 0.692 / 0.692	2/2/2
Transformer	FD002	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6427	99.5	0.643 / 0.643 / 0.643	1/1/1
Transformer	FD002	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	3/3/3
Transformer	FD002	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	4/4/4
Transformer	FD002	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6443	99.6	0.644 / 0.644 / 0.644	5/5/5
Transformer	FD002	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6427	99.5	0.643 / 0.643 / 0.643	2/2/2
Transformer	FD003	Clean	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Clean	10	Fixed var.	0.0000 / 0.000	0.0100	34.0	0.010 / 0.010 / 0.010	4/4/4
Transformer	FD003	Clean	10	MC Drop.	0.0000 / 0.000	0.0100	34.0	0.010 / 0.010 / 0.010	5/5/5
Transformer	FD003	Clean	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Clean	10	CP-norm	0.0000 / 0.000	0.0060	51.0	0.006 / 0.006 / 0.006	3/3/3
Transformer	FD003	Clean	20	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Clean	20	Fixed var.	0.0000 / 0.000	0.0180	48.3	0.018 / 0.018 / 0.018	5/5/5
Transformer	FD003	Clean	20	MC Drop.	0.0000 / 0.000	0.0140	49.0	0.014 / 0.014 / 0.014	3/3/3
Transformer	FD003	Clean	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Clean	20	CP-norm	0.0000 / 0.000	0.0160	68.1	0.016 / 0.016 / 0.016	4/4/4
Transformer	FD003	Clean	30	Heterosc.	0.0000 / 0.000	0.0220	57.2	0.022 / 0.022 / 0.022	1/1/1
Transformer	FD003	Clean	30	Fixed var.	0.0000 / 0.000	0.0320	57.4	0.032 / 0.032 / 0.032	3/3/3
Transformer	FD003	Clean	30	MC Drop.	0.0000 / 0.000	0.0320	57.4	0.032 / 0.032 / 0.032	4/4/4
Transformer	FD003	Clean	30	Ensemble	0.0000 / 0.000	0.0300	59.3	0.030 / 0.030 / 0.030	2/2/2
Transformer	FD003	Clean	30	CP-norm	0.0000 / 0.000	0.0700	62.2	0.070 / 0.070 / 0.070	5/5/5
Transformer	FD003	Noise 1%	10	Heterosc.	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Noise 1%	10	Fixed var.	0.0000 / 0.000	0.0100	34.0	0.010 / 0.010 / 0.010	4/4/4
Transformer	FD003	Noise 1%	10	MC Drop.	0.0000 / 0.000	0.0100	34.0	0.010 / 0.010 / 0.010	5/5/5

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
Transformer	FD003	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Noise 1%	10	CP-norm	0.0000 / 0.000	0.0060	52.7	0.006 / 0.006 / 0.006	3/3/3
Transformer	FD003	Noise 1%	20	Heterosc.	0.0000 / 0.000	0.0004	54.0	0.000 / 0.000 / 0.000	2/2/2
Transformer	FD003	Noise 1%	20	Fixed var.	0.0000 / 0.000	0.0156	49.0	0.016 / 0.016 / 0.016	5/5/5
Transformer	FD003	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0128	50.5	0.013 / 0.013 / 0.013	3/3/3
Transformer	FD003	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0148	66.5	0.015 / 0.015 / 0.015	4/4/4
Transformer	FD003	Noise 1%	30	Heterosc.	0.0000 / 0.000	0.0224	59.3	0.022 / 0.022 / 0.022	1/1/1
Transformer	FD003	Noise 1%	30	Fixed var.	0.0000 / 0.000	0.0332	57.2	0.033 / 0.033 / 0.033	2/2/2
Transformer	FD003	Noise 1%	30	MC Drop.	0.0000 / 0.000	0.0332	57.2	0.033 / 0.033 / 0.033	3/3/3
Transformer	FD003	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0340	58.7	0.034 / 0.034 / 0.034	4/4/4
Transformer	FD003	Noise 1%	30	CP-norm	0.0000 / 0.000	0.0676	62.7	0.068 / 0.068 / 0.068	5/5/5
Transformer	FD003	Bias 5%	10	Heterosc.	0.0004 / 0.005	0.0020	34.0	0.004 / 0.010 / 0.042	2/2/4
Transformer	FD003	Bias 5%	10	Fixed var.	0.0000 / 0.000	0.0132	37.3	0.013 / 0.013 / 0.013	5/4/3
Transformer	FD003	Bias 5%	10	MC Drop.	0.0000 / 0.000	0.0128	37.3	0.013 / 0.013 / 0.013	3/3/2
Transformer	FD003	Bias 5%	10	Ensemble	0.0000 / 0.000	0.0000	–	0.000 / 0.000 / 0.000	1/1/1
Transformer	FD003	Bias 5%	10	CP-norm	0.0008 / 0.010	0.0088	52.6	0.013 / 0.025 / 0.089	4/5/5
Transformer	FD003	Bias 5%	20	Heterosc.	0.0000 / 0.000	0.0052	61.6	0.005 / 0.005 / 0.005	1/1/1
Transformer	FD003	Bias 5%	20	Fixed var.	0.0000 / 0.000	0.0396	49.8	0.040 / 0.040 / 0.040	5/5/5
Transformer	FD003	Bias 5%	20	MC Drop.	0.0000 / 0.000	0.0348	49.7	0.035 / 0.035 / 0.035	4/4/4
Transformer	FD003	Bias 5%	20	Ensemble	0.0000 / 0.000	0.0060	52.0	0.006 / 0.006 / 0.006	2/2/2
Transformer	FD003	Bias 5%	20	CP-norm	0.0000 / 0.000	0.0172	65.3	0.017 / 0.017 / 0.017	3/3/3
Transformer	FD003	Bias 5%	30	Heterosc.	0.0000 / 0.000	0.0380	60.2	0.038 / 0.038 / 0.038	1/1/1
Transformer	FD003	Bias 5%	30	Fixed var.	0.0000 / 0.000	0.0456	58.0	0.046 / 0.046 / 0.046	3/3/3
Transformer	FD003	Bias 5%	30	MC Drop.	0.0000 / 0.000	0.0444	57.8	0.044 / 0.044 / 0.044	2/2/2
Transformer	FD003	Bias 5%	30	Ensemble	0.0000 / 0.000	0.0600	57.9	0.060 / 0.060 / 0.060	4/4/4
Transformer	FD003	Bias 5%	30	CP-norm	0.0004 / 0.002	0.0764	65.9	0.078 / 0.084 / 0.116	5/5/5
Transformer	FD003	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.0460	47.3	0.046 / 0.046 / 0.046	1/1/1
Transformer	FD003	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.2040	52.4	0.204 / 0.204 / 0.204	5/5/5
Transformer	FD003	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.1920	51.4	0.192 / 0.192 / 0.192	4/4/4
Transformer	FD003	Drift 5%	10	Ensemble	0.0000 / 0.000	0.0500	46.0	0.050 / 0.050 / 0.050	2/2/2
Transformer	FD003	Drift 5%	10	CP-norm	0.0000 / 0.000	0.0820	49.2	0.082 / 0.082 / 0.082	3/3/3
Transformer	FD003	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.1740	57.0	0.174 / 0.174 / 0.174	1/1/1
Transformer	FD003	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.3480	69.9	0.348 / 0.348 / 0.348	5/5/5
Transformer	FD003	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.3260	67.6	0.326 / 0.326 / 0.326	4/4/4
Transformer	FD003	Drift 5%	20	Ensemble	0.0000 / 0.000	0.2000	58.2	0.200 / 0.200 / 0.200	3/3/3
Transformer	FD003	Drift 5%	20	CP-norm	0.0000 / 0.000	0.1980	60.2	0.198 / 0.198 / 0.198	2/2/2
Transformer	FD003	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.3060	76.0	0.306 / 0.306 / 0.306	1/1/1
Transformer	FD003	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.4380	85.0	0.438 / 0.438 / 0.438	4/4/4
Transformer	FD003	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.4440	84.9	0.444 / 0.444 / 0.444	5/5/5
Transformer	FD003	Drift 5%	30	Ensemble	0.0000 / 0.000	0.4100	81.7	0.410 / 0.410 / 0.410	3/3/3
Transformer	FD003	Drift 5%	30	CP-norm	0.0000 / 0.000	0.3420	79.6	0.342 / 0.342 / 0.342	2/2/2
Transformer	FD004	Clean	10	Heterosc.	0.0000 / 0.000	0.0304	42.7	0.030 / 0.030 / 0.030	1/1/1
Transformer	FD004	Clean	10	Fixed var.	0.0000 / 0.000	0.0540	40.5	0.054 / 0.054 / 0.054	4/4/4
Transformer	FD004	Clean	10	MC Drop.	0.0000 / 0.000	0.0430	40.4	0.043 / 0.043 / 0.043	2/2/2
Transformer	FD004	Clean	10	Ensemble	0.0000 / 0.000	0.0464	44.5	0.046 / 0.046 / 0.046	3/3/3
Transformer	FD004	Clean	10	CP-norm	0.0000 / 0.000	0.0557	44.1	0.056 / 0.056 / 0.056	5/5/5
Transformer	FD004	Clean	20	Heterosc.	0.0000 / 0.000	0.0287	64.1	0.029 / 0.029 / 0.029	1/1/1
Transformer	FD004	Clean	20	Fixed var.	0.0000 / 0.000	0.0405	58.0	0.041 / 0.041 / 0.041	4/4/4
Transformer	FD004	Clean	20	MC Drop.	0.0000 / 0.000	0.0312	56.7	0.031 / 0.031 / 0.031	2/2/2
Transformer	FD004	Clean	20	Ensemble	0.0000 / 0.000	0.0338	62.8	0.034 / 0.034 / 0.034	3/3/3
Transformer	FD004	Clean	20	CP-norm	0.0000 / 0.000	0.0565	66.0	0.057 / 0.057 / 0.057	5/5/5
Transformer	FD004	Clean	30	Heterosc.	0.0000 / 0.000	0.0473	75.2	0.047 / 0.047 / 0.047	4/2/2
Transformer	FD004	Clean	30	Fixed var.	0.0008 / 0.004	0.0422	70.5	0.046 / 0.059 / 0.127	2/4/4
Transformer	FD004	Clean	30	MC Drop.	0.0008 / 0.004	0.0371	70.5	0.041 / 0.054 / 0.122	1/3/3

Backbone	Data	Cond.	L	Mech.	unrec. (ovr / cond.)	prem.	RUL@t	cost 5/20/100	rank
Transformer	FD004	Clean	30	Ensemble	0.0000 / 0.000	0.0464	71.3	0.046 / 0.046 / 0.046	3/1/1
Transformer	FD004	Clean	30	CP-norm	0.0008 / 0.004	0.0810	78.6	0.085 / 0.098 / 0.165	5/5/5
Transformer	FD004	Noise 1%	10	Heterosc.	0.0008 / 0.011	0.0378	45.4	0.042 / 0.055 / 0.122	1/2/5
Transformer	FD004	Noise 1%	10	Fixed var.	0.0003 / 0.004	0.0700	45.2	0.072 / 0.077 / 0.104	5/5/4
Transformer	FD004	Noise 1%	10	MC Drop.	0.0003 / 0.004	0.0522	42.3	0.054 / 0.059 / 0.086	3/3/2
Transformer	FD004	Noise 1%	10	Ensemble	0.0000 / 0.000	0.0489	45.1	0.049 / 0.049 / 0.049	2/1/1
Transformer	FD004	Noise 1%	10	CP-norm	0.0003 / 0.004	0.0638	50.5	0.065 / 0.071 / 0.098	4/4/3
Transformer	FD004	Noise 1%	20	Heterosc.	0.0002 / 0.001	0.0393	69.9	0.040 / 0.043 / 0.056	1/1/3
Transformer	FD004	Noise 1%	20	Fixed var.	0.0005 / 0.003	0.0562	64.3	0.059 / 0.066 / 0.107	4/4/5
Transformer	FD004	Noise 1%	20	MC Drop.	0.0000 / 0.000	0.0435	61.4	0.044 / 0.044 / 0.044	2/2/1
Transformer	FD004	Noise 1%	20	Ensemble	0.0000 / 0.000	0.0506	67.7	0.051 / 0.051 / 0.051	3/3/2
Transformer	FD004	Noise 1%	20	CP-norm	0.0000 / 0.000	0.0704	71.8	0.070 / 0.070 / 0.070	5/5/4
Transformer	FD004	Noise 1%	30	Heterosc.	0.0005 / 0.002	0.0707	78.6	0.073 / 0.081 / 0.121	2/2/2
Transformer	FD004	Noise 1%	30	Fixed var.	0.0030 / 0.013	0.0646	78.0	0.080 / 0.125 / 0.368	4/5/5
Transformer	FD004	Noise 1%	30	MC Drop.	0.0030 / 0.013	0.0542	76.4	0.069 / 0.115 / 0.358	1/3/4
Transformer	FD004	Noise 1%	30	Ensemble	0.0000 / 0.000	0.0743	77.9	0.074 / 0.074 / 0.074	3/1/1
Transformer	FD004	Noise 1%	30	CP-norm	0.0008 / 0.004	0.1067	82.5	0.111 / 0.124 / 0.191	5/4/3
Transformer	FD004	Bias 5%	10	Heterosc.	0.0093 / 0.122	0.3200	87.0	0.366 / 0.506 / 1.248	1/2/4
Transformer	FD004	Bias 5%	10	Fixed var.	0.0076 / 0.100	0.3661	87.1	0.404 / 0.518 / 1.126	4/4/2
Transformer	FD004	Bias 5%	10	MC Drop.	0.0103 / 0.136	0.3526	87.1	0.404 / 0.558 / 1.382	5/5/5
Transformer	FD004	Bias 5%	10	Ensemble	0.0034 / 0.044	0.3612	87.5	0.378 / 0.429 / 0.699	2/1/1
Transformer	FD004	Bias 5%	10	CP-norm	0.0084 / 0.111	0.3409	86.8	0.383 / 0.510 / 1.185	3/3/3
Transformer	FD004	Bias 5%	20	Heterosc.	0.0177 / 0.105	0.3186	95.9	0.407 / 0.673 / 2.091	2/4/4
Transformer	FD004	Bias 5%	20	Fixed var.	0.0157 / 0.093	0.3576	95.8	0.436 / 0.672 / 1.927	4/3/3
Transformer	FD004	Bias 5%	20	MC Drop.	0.0187 / 0.111	0.3512	96.0	0.445 / 0.726 / 2.225	5/5/5
Transformer	FD004	Bias 5%	20	Ensemble	0.0059 / 0.035	0.3494	96.8	0.379 / 0.468 / 0.940	1/1/1
Transformer	FD004	Bias 5%	20	CP-norm	0.0147 / 0.087	0.3377	96.2	0.411 / 0.631 / 1.806	3/2/2
Transformer	FD004	Bias 5%	30	Heterosc.	0.0208 / 0.091	0.3328	100.4	0.437 / 0.748 / 2.409	2/3/4
Transformer	FD004	Bias 5%	30	Fixed var.	0.0196 / 0.086	0.3642	100.0	0.462 / 0.756 / 2.322	4/4/3
Transformer	FD004	Bias 5%	30	MC Drop.	0.0221 / 0.097	0.3624	100.3	0.473 / 0.805 / 2.573	5/5/5
Transformer	FD004	Bias 5%	30	Ensemble	0.0084 / 0.037	0.3612	100.4	0.403 / 0.530 / 1.205	1/1/1
Transformer	FD004	Bias 5%	30	CP-norm	0.0181 / 0.079	0.3468	100.1	0.437 / 0.708 / 2.153	3/2/2
Transformer	FD004	Drift 5%	10	Heterosc.	0.0000 / 0.000	0.4034	78.3	0.403 / 0.403 / 0.403	1/1/1
Transformer	FD004	Drift 5%	10	Fixed var.	0.0000 / 0.000	0.6768	89.1	0.677 / 0.677 / 0.677	5/5/5
Transformer	FD004	Drift 5%	10	MC Drop.	0.0000 / 0.000	0.6439	87.8	0.644 / 0.644 / 0.644	4/4/4
Transformer	FD004	Drift 5%	10	Ensemble	0.0000 / 0.000	0.4557	81.3	0.456 / 0.456 / 0.456	3/3/3
Transformer	FD004	Drift 5%	10	CP-norm	0.0000 / 0.000	0.4498	81.0	0.450 / 0.450 / 0.450	2/2/2
Transformer	FD004	Drift 5%	20	Heterosc.	0.0000 / 0.000	0.5249	93.8	0.525 / 0.525 / 0.525	1/1/1
Transformer	FD004	Drift 5%	20	Fixed var.	0.0000 / 0.000	0.6591	98.0	0.659 / 0.659 / 0.659	5/5/5
Transformer	FD004	Drift 5%	20	MC Drop.	0.0000 / 0.000	0.6430	97.4	0.643 / 0.643 / 0.643	4/4/4
Transformer	FD004	Drift 5%	20	Ensemble	0.0000 / 0.000	0.5738	94.8	0.574 / 0.574 / 0.574	3/3/3
Transformer	FD004	Drift 5%	20	CP-norm	0.0000 / 0.000	0.5603	95.2	0.560 / 0.560 / 0.560	2/2/2
Transformer	FD004	Drift 5%	30	Heterosc.	0.0000 / 0.000	0.6059	100.8	0.606 / 0.606 / 0.606	1/1/1
Transformer	FD004	Drift 5%	30	Fixed var.	0.0000 / 0.000	0.6354	101.6	0.635 / 0.635 / 0.635	4/4/4
Transformer	FD004	Drift 5%	30	MC Drop.	0.0000 / 0.000	0.6329	101.5	0.633 / 0.633 / 0.633	3/3/3
Transformer	FD004	Drift 5%	30	Ensemble	0.0000 / 0.000	0.6371	101.9	0.637 / 0.637 / 0.637	5/5/5
Transformer	FD004	Drift 5%	30	CP-norm	0.0000 / 0.000	0.6135	100.9	0.614 / 0.614 / 0.614	2/2/2

Per-engine bootstrap of the top-ranked mechanism against the other four (v1.8; the top is chosen from the same data, so these intervals are descriptive): number of the 144 comparisons per backbone and sub-dataset whose 95% interval excludes zero.

Backbone	Data	Excluding zero
LSTM	FD001	79 / 144
LSTM	FD002	86 / 144
LSTM	FD003	89 / 144
LSTM	FD004	70 / 144
Transformer	FD001	63 / 144
Transformer	FD002	90 / 144
Transformer	FD003	81 / 144
Transformer	FD004	101 / 144

S5. Output swap, controls and additive-noise attribution (range scale)

Table 14. Output swap at the decision point, heteroscedastic head, $L = 20$ (exploratory). $\Delta\mu$: mean shift in cycles (perturbed minus clean, same model). Premature rates: clean, mean perturbed only, scale perturbed only, both; Δ_σ: effect of the perturbed scale given the perturbed mean (points, with engine-bootstrap interval). Unrecognised rates: mean only, scale only, both.

Cell	Cond.	$\Delta\mu$	Prem. clean	μ only	σ only	both	$\Delta_\sigma \mid \mu_1$	Unrec. μ only	σ only	both
LSTM FD002	Drift	-64.5	3.7	69.4	0.2	69.3	-0.1 [-0.6, +0.4]	0.0	0.2	0.0
LSTM FD002	Bias	+7.3	3.7	16.0	4.1	17.0	+0.9 [+0.7, +1.2]	10.7	0.0	7.8
LSTM FD004	Drift	-49.2	3.9	49.5	1.1	59.0	+9.5 [+6.8, +12.2]	0.0	0.8	0.0
LSTM FD004	Bias	-21.1	3.9	29.8	3.2	30.2	+0.3 [-0.6, +1.3]	4.5	0.6	2.5
Transformer FD002	Drift	-65.7	1.5	68.2	0.5	69.2	+0.9 [+0.3, +1.7]	0.0	0.2	0.0
Transformer FD002	Bias	+11.5	1.5	14.2	2.8	13.8	-0.4 [-0.6, -0.2]	11.3	0.1	10.2
Transformer FD004	Drift	-58.9	2.9	55.2	1.3	52.5	-2.7 [-4.3, -1.3]	0.0	0.0	0.0
Transformer FD004	Bias	-27.9	2.9	33.1	3.7	31.9	-1.2 [-2.2, -0.2]	2.6	0.2	1.8
LSTM FD001	Drift	-19.5	0.0	14.0	0.0	2.6	-11.4 [-16.6, -6.6]	0.0	0.2	0.0
LSTM FD001	Bias	+0.0	0.0	0.9	0.2	0.1	-0.8 [-1.4, -0.4]	0.5	0.2	0.5
LSTM FD003	Drift	-38.6	0.0	38.0	0.0	13.8	-24.2 [-31.2, -17.4]	0.0	0.0	0.0
LSTM FD003	Bias	-3.2	0.0	2.9	0.3	0.6	-2.3 [-3.6, -1.2]	0.1	0.0	0.1
Transformer FD001	Drift	-20.0	0.0	12.8	0.0	2.4	-10.4 [-15.6, -5.8]	0.0	0.0	0.0
Transformer FD001	Bias	+1.3	0.0	0.1	0.2	0.0	-0.1 [-0.2, +0.0]	0.3	0.0	0.2
Transformer FD003	Drift	-41.8	0.0	43.2	0.0	17.4	-25.8 [-32.6, -19.4]	0.0	0.0	0.0
Transformer FD003	Bias	-2.9	0.0	2.4	0.5	0.5	-1.8 [-3.1, -0.8]	0.0	0.0	0.0

Table 15. Output swap for the sign-pattern variants of the 5% shift (Section 3.3), heteroscedastic head, $L = 20$, exploratory, inference only. $\Delta\mu$: mean shift in cycles; σ/σ_0: mean ratio of perturbed to clean scale. Effects are in percentage points with engine-bootstrap 95% intervals: mean effect = perturbed mean with clean scale minus clean; scale effect = both perturbed minus perturbed mean with clean scale.

Cell	Shift	$\Delta\mu$	σ/σ_0	Prem., mean effect	Prem., scale effect	Unrec., mean effect	Unrec., scale effect
LSTM FD001	Ramp (+)	-19.5	1.06	+14.0 [+8.4, +20.0]	-11.4 [-16.8, -6.6]	-0.2 [-0.6, +0.0]	+0.0 [+0.0, +0.0]
LSTM FD001	Const. +	-11.4	1.02	+9.2 [+5.0, +14.2]	-7.4 [-11.8, -3.4]	-0.2 [-0.6, +0.0]	+0.0 [+0.0, +0.0]
LSTM FD001	Const. −	+7.3	0.90	+0.4 [+0.0, +1.0]	-0.4 [-1.0, +0.0]	+2.8 [+0.6, +5.6]	+0.0 [+0.0, +0.0]
LSTM FD002	Ramp (+)	-64.5	0.53	+65.7 [+59.9, +71.2]	-0.1 [-0.6, +0.4]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
LSTM FD002	Const. +	-59.9	0.59	+61.3 [+55.1, +66.7]	-1.6 [-2.5, -0.7]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
LSTM FD002	Const. −	+36.6	1.21	-2.1 [-3.8, -0.7]	+0.6 [+0.2, +1.0]	+17.5 [+13.4, +21.9]	-5.4 [-7.3, -3.6]
LSTM FD003	Ramp (+)	-38.6	0.91	+38.0 [+30.0, +46.4]	-24.2 [-31.2, -18.0]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
LSTM FD003	Const. +	-3.8	1.05	+0.8 [+0.0, +2.0]	-0.8 [-2.0, +0.0]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
LSTM FD003	Const. −	+7.7	0.95	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]	+0.4 [+0.0, +1.2]	+0.0 [+0.0, +0.0]
LSTM FD004	Ramp (+)	-49.2	1.10	+45.7 [+39.8, +51.2]	+9.5 [+6.8, +12.4]	-0.1 [-0.3, +0.0]	+0.0 [+0.0, +0.0]
LSTM FD004	Const. +	-42.5	0.91	+38.5 [+32.3, +44.1]	-8.9 [-11.5, -6.4]	-0.1 [-0.3, +0.0]	+0.0 [+0.0, +0.0]
LSTM FD004	Const. −	+19.1	1.27	-3.0 [-5.0, -1.3]	+2.4 [+1.2, +3.7]	+10.8 [+7.2, +14.6]	-7.2 [-9.8, -4.6]
Transformer FD001	Ramp (+)	-20.0	1.07	+12.8 [+7.8, +18.0]	-10.4 [-15.2, -6.0]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
Transformer FD001	Const. +	-10.6	1.06	+1.4 [+0.2, +3.0]	-1.0 [-2.4, +0.0]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
Transformer FD001	Const. −	+8.4	0.94	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]	+1.4 [+0.0, +3.2]	-0.4 [-1.0, +0.0]
Transformer FD002	Ramp (+)	-65.7	0.57	+66.7 [+61.0, +72.3]	+0.9 [+0.3, +1.7]	-0.1 [-0.2, +0.0]	+0.0 [+0.0, +0.0]
Transformer FD002	Const. +	-65.0	0.57	+65.5 [+59.8, +71.1]	-1.1 [-1.9, -0.4]	-0.1 [-0.2, +0.0]	+0.0 [+0.0, +0.0]
Transformer FD002	Const. −	+42.4	0.90	-1.5 [-2.8, -0.5]	+0.0 [+0.0, +0.0]	+16.0 [+11.8, +20.5]	-1.3 [-2.1, -0.7]
Transformer FD003	Ramp (+)	-41.8	0.80	+43.2 [+34.8, +52.0]	-25.8 [-32.2, -19.2]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
Transformer FD003	Const. +	-11.0	1.02	+6.6 [+3.6, +10.0]	-5.8 [-9.0, -3.0]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
Transformer FD003	Const. −	+9.1	1.02	+0.0 [+0.0, +0.0]	+0.2 [+0.0, +0.6]	+0.4 [+0.0, +1.2]	+0.0 [+0.0, +0.0]
Transformer FD004	Ramp (+)	-58.9	0.78	+52.3 [+46.8, +58.1]	-2.7 [-4.3, -1.2]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
Transformer FD004	Const. +	-49.3	0.89	+40.6 [+34.9, +46.2]	-7.5 [-10.5, -4.8]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
Transformer FD004	Const. −	+5.0	1.19	+3.1 [+1.2, +5.2]	-2.8 [-4.4, -1.4]	+8.1 [+5.3, +11.2]	-0.3 [-0.8, +0.1]

Table 16. Waveform controls at 5% of range, heteroscedastic head, terminal-window PICP (released result files except the constant + offset, which is from the sign-coherent control). Ramp: drift family; reversed, shuffled, per-trajectory (continuous), bias with equal end value and single-channel ramp as in the earlier analysis; $k = 1, 3, 5$: fixed-end shuffled ramp (FD002 only; "–": not available).

Cell	Clean	Ramp	Const. +	Reversed	Shuffled	Per traj.	Bias eq.	Single ch.	$k = 1$	$k = 3$	$k = 5$
LSTM FD001	0.864	0.644	0.750	0.684	0.859	0.752	0.678	0.868	–	–	–
LSTM FD002	0.913	0.048	0.043	0.679	0.285	0.042	0.486	0.629	0.288	0.197	0.127
Transformer FD001	0.880	0.642	0.852	0.764	0.882	0.826	0.883	0.900	–	–	–
Transformer FD002	0.892	0.025	0.024	0.691	0.123	0.024	0.230	0.538	0.071	0.034	0.027

Table 17. Drift controls at 5% of range, FD002, heteroscedastic head: excursion and PICP.

Perturbation	f_{oob}	LSTM	Transformer
Ramp (per window)	0.114	0.048	0.025
Ramp (contin. trajectory)	0.141	0.042	0.024
Bias, equal end value	0.150	0.486	0.230
Reversed ramp	0.111	0.679	0.691
Shuffled ramp	0.112	0.285	0.123
Single-channel ramp (s_9)	7.4e-05	0.629	0.538

Additive-noise attribution (v1.8 tables). Four-way frozen-output decomposition at the point where the heteroscedastic head has lost ten points of coverage. Scale contribution: share of the total ten-point loss (positive: the live scale makes the loss larger), under both decomposition orders.

Table 18. Attribution, LSTM.

		FD001	FD002	FD003	FD004
Heteroscedastic	scale, mean first	+28.3%	+8.9%	+3.6%	−5.8%
	scale, scale first	+37.5%	+30.4%	+27.0%	+36.9%
CP-norm	scale, mean first	+28.4%	+14.3%	+9.8%	−6.4%
	scale, scale first	+35.6%	+28.8%	+28.1%	+33.6%
Mean dominates (point estimate), both orders		yes	yes	yes	yes
CI excludes zero, mean first (het./CP)		yes/yes	yes/yes	yes/yes	yes/yes
CI excludes zero, scale first (het./CP)		yes/yes	yes/yes	yes/yes	yes/yes
Baseline clamp fraction		0.34	0.22	0.40	0.076

Table 19. Attribution, Transformer.

		FD001	FD002	FD003	FD004
Heteroscedastic	scale, mean first	+20.0%	+11.4%	+0.2%	+3.2%
	scale, scale first	+20.3%	+31.0%	+21.0%	+21.9%
CP-norm	scale, mean first	+17.7%	+11.6%	+6.5%	+4.3%
	scale, scale first	+24.9%	+32.5%	+16.8%	+23.7%
Mean dominates (point estimate), both orders		yes	yes	yes	yes
CI excludes zero, mean first (het./CP)		yes/yes	yes/yes	yes/yes	yes/yes
CI excludes zero, scale first (het./CP)		yes/yes	yes/yes	yes/yes	yes/yes

Table 20. Four-way attribution values; C_{ij}: coverage with mean from condition i and scale from condition j (0 clean, 1 perturbed); order A varies the mean first.

Backbone	Data	Mech.	f_{oob} (%)	C_{00}	C_{10}	C_{01}	C_{11}	μ (A)	σ (A)	μ (B)	σ (B)	Int.	Both
LSTM	FD001	Heterosc.	0.55	0.864	0.792	0.826	0.764	-0.0717	-0.0283	-0.0625	-0.0375	+0.0092	yes
LSTM	FD001	CP-norm	0.85	0.878	0.806	0.842	0.778	-0.0716	-0.0284	-0.0644	-0.0356	+0.0072	yes
LSTM	FD002	Heterosc.	1.04	0.913	0.822	0.883	0.813	-0.0911	-0.0089	-0.0696	-0.0304	+0.0215	yes
LSTM	FD002	CP-norm	0.99	0.932	0.846	0.903	0.832	-0.0857	-0.0143	-0.0712	-0.0288	+0.0145	yes
LSTM	FD003	Heterosc.	0.24	0.858	0.762	0.831	0.758	-0.0964	-0.0036	-0.0730	-0.0270	+0.0234	yes
LSTM	FD003	CP-norm	0.63	0.896	0.806	0.868	0.796	-0.0902	-0.0098	-0.0719	-0.0281	+0.0183	yes
LSTM	FD004	Heterosc.	1.40	0.885	0.779	0.848	0.785	-0.1058	+0.0058	-0.0631	-0.0369	+0.0427	yes
LSTM	FD004	CP-norm	1.42	0.877	0.771	0.843	0.777	-0.1064	+0.0064	-0.0664	-0.0336	+0.0400	yes
Transformer	FD001	Heterosc.	0.89	0.880	0.800	0.860	0.780	-0.0800	-0.0200	-0.0797	-0.0203	+0.0002	yes
Transformer	FD001	CP-norm	0.63	0.892	0.810	0.867	0.792	-0.0823	-0.0177	-0.0751	-0.0249	+0.0071	yes
Transformer	FD002	Heterosc.	0.95	0.892	0.803	0.861	0.792	-0.0886	-0.0114	-0.0690	-0.0310	+0.0195	yes
Transformer	FD002	CP-norm	0.95	0.911	0.823	0.879	0.811	-0.0884	-0.0116	-0.0675	-0.0325	+0.0210	yes
Transformer	FD003	Heterosc.	0.11	0.870	0.770	0.849	0.770	-0.0998	-0.0002	-0.0790	-0.0210	+0.0208	yes
Transformer	FD003	CP-norm	0.26	0.876	0.783	0.859	0.776	-0.0935	-0.0065	-0.0832	-0.0168	+0.0102	yes
Transformer	FD004	Heterosc.	0.54	0.842	0.745	0.820	0.742	-0.0968	-0.0032	-0.0781	-0.0219	+0.0187	yes
Transformer	FD004	CP-norm	0.53	0.850	0.755	0.827	0.750	-0.0957	-0.0043	-0.0763	-0.0237	+0.0194	yes

Table 21. Per-seed paired bootstrap of $|\Delta_\mu| - |\Delta_\sigma|$ (five seeds, 2000 resamples; conditional on the test set, the seeded realisations and the interpolated crossover).

Backbone	Data	Mech.	Mean first: mean [95% CI]	> 0	Scale first: mean [95% CI]	> 0
LSTM	FD001	Heterosc.	+0.043 [+0.029, +0.058]	yes	+0.025 [+0.009, +0.039]	yes
LSTM	FD001	CP-norm	+0.043 [+0.028, +0.061]	yes	+0.029 [+0.006, +0.051]	yes
LSTM	FD002	Heterosc.	+0.082 [+0.070, +0.094]	yes	+0.039 [+0.028, +0.051]	yes
LSTM	FD002	CP-norm	+0.071 [+0.046, +0.096]	yes	+0.042 [+0.028, +0.057]	yes
LSTM	FD003	Heterosc.	+0.087 [+0.063, +0.113]	yes	+0.043 [+0.032, +0.055]	yes
LSTM	FD003	CP-norm	+0.078 [+0.039, +0.116]	yes	+0.044 [+0.027, +0.061]	yes
LSTM	FD004	Heterosc.	+0.096 [+0.086, +0.107]	yes	+0.029 [+0.017, +0.046]	yes
LSTM	FD004	CP-norm	+0.100 [+0.084, +0.120]	yes	+0.028 [+0.020, +0.037]	yes
Transformer	FD001	Heterosc.	+0.060 [+0.036, +0.084]	yes	+0.058 [+0.052, +0.062]	yes
Transformer	FD001	CP-norm	+0.065 [+0.036, +0.095]	yes	+0.048 [+0.030, +0.066]	yes
Transformer	FD002	Heterosc.	+0.077 [+0.065, +0.092]	yes	+0.038 [+0.031, +0.045]	yes
Transformer	FD002	CP-norm	+0.077 [+0.060, +0.093]	yes	+0.035 [+0.028, +0.041]	yes
Transformer	FD003	Heterosc.	+0.087 [+0.049, +0.121]	yes	+0.049 [+0.033, +0.066]	yes
Transformer	FD003	CP-norm	+0.083 [+0.041, +0.126]	yes	+0.066 [+0.052, +0.081]	yes
Transformer	FD004	Heterosc.	+0.108 [+0.093, +0.124]	yes	+0.041 [+0.025, +0.055]	yes
Transformer	FD004	CP-norm	+0.098 [+0.087, +0.111]	yes	+0.046 [+0.029, +0.060]	yes

S6. Sensitivity and scope checks

Evaluated engines. FD001–FD004: 100, 253, 100 and 237 evaluated against 100, 259, 100 and 248 in the test files; engines with fewer than 30 recorded cycles cannot form a window. **Label instant.** Scoring against the current-cycle label (official RUL, clipped) instead of $\min(\text{RUL} - 1, 125)$ changes any decision rate of any mechanism, condition, lead time or backbone by at most 2.0 percentage points; for the headline drift and bias cells it changes them by at most 1.3 points. **Operating settings.** Perturbing the three operating-setting channels as well as the sensors raised the premature rate under drift on FD004 from 0.52–0.59 to 0.67–0.68; the settings alone gave 0.12–0.23; on FD002 0.69 against 0.70. **Agreement with the release.** The per-engine re-derivation reproduced 960 released decision rates with maximum absolute difference 1e-16, and the released ramp coverage for all eight backbone–sub-dataset cells (`coherent_bias_control.json`).

Table 22. Sign-coherent shift control, heteroscedastic head, 5% of range: PICP / premature (%) / unrecognised (%), $L = 20$ (exploratory).

	Clean PICP	Ramp (+)	Constant $+5\%$	Constant -5%
LSTM FD001	0.86	0.64 / 3 / 0.0	0.75 / 2 / 0.0	0.65 / 0 / 3.0
LSTM FD002	0.91	0.05 / 69 / 0.0	0.04 / 63 / 0.0	0.31 / 2 / 12.2
LSTM FD003	0.86	0.21 / 14 / 0.0	0.82 / 0 / 0.0	0.67 / 0 / 0.4
LSTM FD004	0.88	0.31 / 59 / 0.0	0.35 / 33 / 0.0	0.66 / 3 / 3.7
Transformer FD001	0.88	0.64 / 2 / 0.0	0.85 / 0 / 0.0	0.74 / 0 / 1.0
Transformer FD002	0.89	0.03 / 69 / 0.0	0.02 / 66 / 0.0	0.28 / 0 / 14.8
Transformer FD003	0.87	0.10 / 17 / 0.0	0.69 / 1 / 0.0	0.74 / 0 / 0.4
Transformer FD004	0.86	0.05 / 52 / 0.0	0.10 / 36 / 0.0	0.57 / 3 / 7.8

S7. Provenance, latency and reference verification

Table 23. Provenance of every v1.8 table: script or result file, label version, evaluation unit, engine weighting, aggregation, perturbation-scale reference.

Table	Script / result file	Labels	Unit	Eng.-wt.	Aggregation	Noise range
Protocol 2x2 (Table I)	build_table1.py (re-inference over all four cells' existing checkpoints under this release's protocol); underlying checkpoints from train_lstm.py (V/F) and protocol_2x2_quadrant4 .py (T/F); the T/W' and V/W' cells' checkpoints come from protocol_ 2x2_controlled_retrain.py (retrained with the same 60% canonical fit_ units as V/F/T/F, isolating selection-criterion and scaler-scope as the only remaining variables), shipped under results/checkpoints/lstm_2x2_ controlled/; the original, uncontrolled T/W/V/W checkpoints and their training scripts are kept for provenance under results/superseded/ + code/superseded/ and are no longer read by this table	min(official_ RUL-1,125)	terminal	yes	mean of 5 seeds; ECE on pooled seeds	–
Clean benchmark (Table II)	eval_nll_clean_b1.py, eval_splitcp_ b1.py, eval_mcdropout_b1.py, eval_ fd003_mechanism_b1.py (re-evaluate existing checkpoints under this release's test-truth convention, no retraining); fair_calibration_main_table .py (fixed variance, MC Dropout); table2_clean_merge.py (others); eval_ *_per_engine_coverage.py (per-engine column); eval_*_deep_ensemble.py (ensemble rows)	min(official_ RUL-1,125)	terminal (per-engine: trajectory)	yes	mean of 5 seeds; ECE per seed	–
Same-partition ensemble (Table III)	eval_samesplit_ensemble_b1.py	min(official_ RUL-1,125)	terminal	yes	one partition (seed 42), five initialisations	–
Attribution, LSTM (Table IV)	attribution_interp_lstm.py; clamp fraction from clamp_frac.py	min(official_ RUL-1,125)	terminal	yes	seed-pooled predictions; interpolated crossover	sensor channels only on FD002/FD004 (15 of 18 feature columns; the 3 operating-condition settings are left unperturbed); official training file for FD001/FD003

Table 23. Provenance of every v1.8 table: script or result file, label version, evaluation unit, engine weighting, aggregation, perturbation-scale reference.

Table	Script / result file	Labels	Unit	Eng.-wt.	Aggregation	Noise range
Structured degradation (Table V)	run_sweep_degradation_transformer .py with sweep_engine.py (both backbones), half_life_degradation.py	min(official_ RUL-1,125)	terminal	yes	5 models x 5 realisations, grand mean	sensor channels only on FD002/FD004 (15 of 18 feature columns); official training file for FD001/FD003. The gain arm's sensor-only restriction is now actually enforced – previously inject_gain_fixed_pct_raw's (1+/-k) multiplier did not depend on the zeroed full_scale_ range used by the other arms, so gain kept perturbing the 3 setting columns even after an earlier sensor-only fix; it now takes an explicit sensor_mask.
Drift controls (Table VI)	drift_controls.py; drift_bias_equal_magnitude_control.py; drift_fixed_endpoint_shuffle_control.py	min(official_ RUL-1,125)	terminal	yes	5 models; deterministic except shuffled/bias sign draws	sensor channels only on FD002/FD004 (15 of 18 feature columns); official training file for FD001/FD003
Decision stress test (Table VII)	maintenance_decision_one_sided.py; unclipped RUL from maintenance_stress_test_rul_correction_v2.py	min(official_ RUL-1,125) for triggers; unclipped RUL reported as official_ RUL-1	terminal	yes	5 models x 5 realisations	sensor channels only on FD002/FD004 (15 of 18 feature columns); official training file for FD001/FD003
Pre-flight diagnostics (not in main tables)	diagnostic_perturbation_target_a1.py; diagnostic_target_alignment_a2.py (see results/diagnostics/)	both official_ RUL and min(official_ RUL-1,125) compared side by side (A2); sensors-only/settings-only/joint compared side by side (A1)	terminal	yes	5 seeds	A1 compares sensor-only vs settings-only vs joint on FD002/FD004; A2 does not perturb features
Fair calibration (Appendix B)	fair_calibration_sigma_fixed.py	min(official_ RUL-1,125)	terminal	yes	mean of 5 seeds	–

Table 23. Provenance of every v1.8 table: script or result file, label version, evaluation unit, engine weighting, aggregation, perturbation-scale reference.

Table	Script / result file	Labels	Unit	Eng.-wt.	Aggregation	Noise range
Attribution, Transformer (Appendix C)	attribution_interp_transformer.py	min(official_RUL-1,125)	terminal	yes	seed-pooled predictions; interpolated crossover	sensor channels only on FD002/FD004 (15 of 18 feature columns); official training file for FD001/FD003
Seeds and ensemble size (Appendix D)	eval_lstm_ensemble_n5_vs_n15.py; ensemble_size_and_independent_replication.py	min(official_RUL-1,125)	terminal	yes	15 seeds; three disjoint ensembles	–
Latency (Appendix E)	latency_lstm.py; latency_transformer.py	–	–	–	median and IQR over 11 repetitions	–

New analyses of this revision: `code_jrr/jrr_decision_per_engine.py` (inference only; per-engine lower bounds), `code_jrr/jrr_derived_stats.py` (engine bootstrap, output swap, label-instant sensitivity, rank stability, f_{oob} points), `code_jrr/jrr_coherent_bias_control.py` (sign-coherent shift control), `code_jrr/jrr_coherent_swap.py` (output swap for the sign-pattern variants), `code_jrr/jrr_clean_same_path.py` (clean baseline under the stress-test processing) and `code_jrr/jrr_clip_sensitivity.py` (clipping sensitivity). The CSV files `decision_engine_bootstrap_all.csv`, `rank_stability_all96.csv` and `clean_same_path_table.csv` in `results_jrr/derived/` enumerate the corresponding tables in full. Scripts of the SD-scale revision (`jrr_revision/code/` in the project repository; inference from the released checkpoints only, no training): `rev_generate.py` and `rev_generate_sd.py` (range- and SD-scale perturbations, calibration windows, trajectory outputs), `rev_analyse.py` and `rev_analyse_sd.py` (calibration units, condition tables, decision-sensitivity grid, ranking regret, trajectory first trigger), `rev_perchannel.py` (single-channel ramps), `rev_traj_sd.py` (first trigger along test trajectories), `rev_figs.py` (Figures 2 and 3), `make_rev_numbers.py` and `make_supp_tables.py` (every number and table of the main text and of S9–S11). Outputs are under `results_jrr/derived/`; `results/` is unchanged from the original release (tag `v1.8-submission`).

Latency (v1.8 measurement, not re-run). Per-sample latency in microseconds, median (IQR) over eleven repetitions, one NVIDIA RTX A5000 Laptop GPU, CUDA-event timing; columns: heteroscedastic, fixed variance, CP-norm, MC Dropout ($T = 50$), ensemble ($M = 5$).

Data	Backbone	Batch	Heterosc.	Fixed var.	CP-norm	MC Drop.	Ensemble
FD001	LSTM	1	553.89 (71.65)	402.37 (24.66)	527.46 (29.74)	13801.47 (1039.97)	1761.28 (91.54)
FD001	LSTM	32	10.98 (1.04)	9.69 (0.32)	10.56 (0.97)	436.10 (16.21)	44.87 (1.34)
FD001	LSTM	512	1.80 (0.12)	1.68 (0.08)	1.72 (0.08)	89.64 (1.10)	8.02 (0.22)
FD001	Transformer	1	802.82 (67.09)	717.82 (37.25)	798.66 (43.89)	48316.42 (1233.41)	3516.42 (57.34)
FD001	Transformer	32	21.98 (1.88)	20.61 (1.22)	23.20 (1.21)	1940.29 (64.85)	113.95 (3.30)
FD001	Transformer	512	5.22 (0.09)	5.69 (0.11)	5.26 (0.05)	420.23 (0.40)	35.33 (0.18)
FD002	LSTM	1	364.54 (46.51)	324.54 (32.78)	366.59 (28.05)	15504.38 (355.22)	1605.63 (78.40)
FD002	LSTM	32	12.77 (3.09)	10.50 (1.09)	11.93 (0.61)	498.24 (11.05)	56.42 (1.86)
FD002	LSTM	512	2.80 (0.06)	2.73 (0.04)	2.78 (0.07)	140.70 (0.81)	13.23 (0.07)
FD002	Transformer	1	827.39 (76.24)	761.86 (87.02)	812.03 (98.37)	58784.77 (1043.97)	3904.51 (136.19)
FD002	Transformer	32	25.63 (2.13)	24.96 (2.18)	25.38 (1.40)	2109.92 (82.58)	127.04 (3.76)
FD002	Transformer	512	27.94 (1.07)	25.65 (1.24)	28.07 (0.54)	442.88 (40.97)	36.61 (0.04)
FD003	LSTM	1	351.23 (13.31)	318.46 (20.48)	335.87 (29.70)	14606.34 (1037.82)	1685.50 (80.42)
FD003	LSTM	32	12.13 (1.17)	9.50 (0.44)	11.55 (0.58)	487.29 (10.01)	55.74 (3.15)
FD003	LSTM	512	2.81 (0.04)	2.80 (0.05)	2.90 (0.06)	140.14 (0.48)	13.07 (0.20)
FD003	Transformer	1	755.71 (126.98)	681.09 (123.94)	739.33 (58.82)	57734.14 (2224.64)	3513.34 (39.94)
FD003	Transformer	32	25.32 (1.04)	22.21 (1.38)	25.38 (1.14)	1828.22 (58.99)	105.38 (3.37)
FD003	Transformer	512	7.06 (0.05)	7.04 (0.05)	7.16 (0.11)	418.41 (0.56)	35.34 (0.34)
FD004	LSTM	1	355.33 (24.03)	282.62 (9.30)	347.14 (13.79)	15542.27 (262.64)	1697.79 (80.74)
FD004	LSTM	32	11.07 (1.14)	8.61 (0.64)	10.78 (1.15)	493.25 (17.49)	51.46 (2.17)
FD004	LSTM	512	2.82 (0.08)	2.79 (0.08)	2.80 (0.08)	139.76 (0.96)	12.97 (0.14)
FD004	Transformer	1	887.81 (86.02)	786.43 (84.18)	798.69 (46.51)	57554.94 (26801.15)	4062.21 (237.57)
FD004	Transformer	32	25.25 (2.43)	20.49 (0.99)	22.18 (0.42)	2005.98 (124.10)	117.09 (3.34)
FD004	Transformer	512	6.15 (0.11)	6.34 (0.29)	6.27 (0.17)	419.99 (0.38)	35.56 (0.10)

Reference verification. Each reference of the revised manuscript that carries a DOI was checked against Crossref (title, author order, journal, volume, article number and year) or, for the preprint, the arXiv abstract page, on 30 October 2026. References without DOI were not machine-checked.

#	Key	Check
1	kamariotis2024	Crossref: title, authors, journal, volume, number and year matched (2026-10-30)
2	mitici2023	Crossref: title, authors, journal, volume, number and year matched (2026-10-30)
3	depater2022	Crossref: title, authors, journal, volume, number and year matched (2026-10-30)
4	lee2022	Crossref: title, authors, journal, volume, number and year matched (2026-10-30)
5	basora2025	Crossref: title, authors, journal, volume, number and year matched (2026-10-30)
6	piao2025	Crossref: title, authors, journal, volume, number and year matched (2026-10-30)
7	xu2026	Crossref: title, authors, journal, volume, number and year matched (2026-10-30)
8	javanmardi2022	arXiv abstract page: title, authors, journal, volume, number and year matched (2026-10-30)
9	hachem2024	Crossref: title, authors, journal, volume, number and year matched (2026-10-30)
10	xu2023	Crossref: title, authors, journal, volume, number and year matched (2026-10-30)
11	tang2026	Crossref: title, authors, journal, volume, number and year matched (2026-10-30)
12	saxena2008	no DOI; not machine-checked
13	nix1994	Crossref: title, authors, journal, volume, number and year matched (2026-10-30)
14	gal2016	no DOI; not machine-checked
15	kendall2017	no DOI; not machine-checked
16	lakshminarayanan2017	no DOI; not machine-checked
17	barber2023	no DOI; not machine-checked

S8. Comparison with the nearest prior work

The search behind this table combined Consensus (three queries, October 2026: interval RUL evaluated through a maintenance rule; RUL uncertainty under distribution shift or sensor faults; decision-oriented evaluation of probabilistic prognostics) with the reference lists of the retrieved papers. Papers marked F were read in full text (author or repository version); papers marked A were assessed from the abstract and, where available, the publisher's summary. Columns: *Interval* – a prediction interval or RUL distribution is evaluated; *Sensor shift* – structured sensor perturbation (drift, bias, gain, failure) is applied at test time; *Rule* – the interval or distribution feeds an explicit maintenance rule; *Both errors* – premature and late maintenance outcomes are reported separately; *Engine CI* – uncertainty is resampled at the unit level.

Table 26. Nearest prior work and what this study adds. ✓: present; (✓): partly; –: absent.

Study	Read	Interval	Sensor shift	Rule	Both errors	Engine CI	Relation to this study
Kamariotis et al. 2024 (RESS 242)	F	(✓)	–	✓	(✓)	–	Decision-oriented metric (long-run cost rate relative to perfect prognostics) on FD001, 20 evaluation units, sequential policies; no sensor degradation. Closest in spirit: evaluates prognostics through decisions. This study asks how sensor degradation changes that evaluation, which their metric does not address.
Basora et al. 2025 (RESS 253)	F	✓	(✓)	–	–	–	UQ benchmark (HNN, MCD, DE, BNNs) on N-CMAPSS incl. OOD units; calibration/sharpness only; "no method clearly outperforms the others". No maintenance rule, no sensor faults.
Mitici et al. 2023 (RESS 234)	F	✓	–	✓	(✓)	–	MC-dropout RUL distribution in a renewal-reward replacement model, all four sub-datasets; optimal replacement close to the 99% lower bound. Clean inputs only; motivates our use of a lower-bound rule and of alternative bound levels (S9).
Lee & Mitici 2023 (RESS 230)	A	✓	–	✓	(✓)	–	DRL maintenance from probabilistic RUL; clean inputs.
de Pater et al. 2022 (RESS 221)	A	(✓)	–	✓	(✓)	–	Alarm-based scheduling with safety factor; clean inputs.
Hachem et al. 2024 (RESS 243)	A	(✓)	✓	–	–	–	Joint system–sensor degradation (Wiener/Gamma) with particle filters; simulated; no learned intervals or maintenance rule.
Li/Xu et al. 2023 (JAS 10); Tang 2026 (PLoS One); Zhang et al. 2025 (TIM)	A	–	✓	–	–	–	Robust point RUL under sensor malfunction or missing sensors; accuracy metrics only.
Javanmardi & Hüllermeier 2022 (arXiv); Piao et al. 2025 (TIM); Xu et al. 2026 (RESS 275)	A	✓	–	–	–	–	Conformal RUL intervals on C-MAPSS; coverage/width; exchangeability assumed, calibration and test sampling units not compared.
Moccardi et al. 2025 (Future Internet)	A	✓	(✓)	–	–	–	Naive, weighted and bootstrap conformal on CMAPSS; recommends weighted conformal under non-exchangeability; no rule.
Robinson 2026 (IJPHM 17)	A	✓	–	(✓)	(✓)	–	Asymmetric CQR that allocates miscoverage towards the late side; clean inputs.
Alomari 2026 (IEEE Access)	A	✓	(✓)	✓	(✓)	–	Multi-task conformal model on N-CMAPSS with OOD units and replacement policy; OOD = other units/conditions, not sensor degradation.
Zhu et al. 2025 (QREI 41); Chen et al. 2022 (Neurocomputing)	A	✓	–	✓	–	–	Interval/distribution RUL into cost-rate maintenance optimisation; clean inputs.
This study		✓	✓	✓	✓	✓	Lower-bound rule evaluated under structured sensor degradation with both trigger errors, engine-level intervals, output swaps, sign-pattern, channel-subset and trajectory controls.

Read in full text: Kamariotis et al. (arXiv:2306.03759), Basora et al. (arXiv:2302.04730), Mitici et al. (TU Delft repository version). The others were assessed from abstracts; their entries describe only what the abstracts state.

S9. Calibration unit and one-sided coverage on clean inputs

The conformal quantile and the fixed residual scales are computed from all overlapping windows of the calibration engines, whereas the test unit is one terminal window per engine. Table 27 gives the one-sided coverage of the 95% lower bound on clean inputs for all mechanisms (engine-bootstrap 95% interval for the heteroscedastic head and CP-norm) and the one-sided coverage of CP-norm when the conformal quantile is recomputed with three alternative calibration units: each calibration engine weighted equally (engine-weighted), calibration windows reweighted to the histogram of clipped test labels (label-matched; it uses the test label distribution and is a diagnostic, not a deployable procedure) and one randomly drawn window per calibration engine (median over 500 draws).

Table 27. One-sided coverage on clean inputs by mechanism and calibration unit (mean of five seeds).

		One-sided coverage of the 95% lower bound, clean					CP-norm, other calibration units		
Backbone	Data	Fixed var.	Heterosc.	MC Drop.	Ensemble	CP-norm	Engine-wt.	Label-matched	One/engine
LSTM	FD001	0.914	0.904 [0.85, 0.95]	0.928	0.940	0.938 [0.90, 0.97]	0.948	0.950	0.956
LSTM	FD002	0.958	0.947 [0.93, 0.97]	0.965	0.972	0.955 [0.94, 0.97]	0.966	0.957	0.990
LSTM	FD003	0.904	0.888 [0.83, 0.94]	0.914	0.930	0.924 [0.88, 0.96]	0.942	0.944	0.946
LSTM	FD004	0.938	0.916 [0.89, 0.94]	0.947	0.941	0.940 [0.92, 0.96]	0.960	0.961	0.988
Transformer	FD001	0.908	0.926 [0.88, 0.96]	0.950	0.950	0.952 [0.91, 0.98]	0.960	0.958	0.958
Transformer	FD002	0.961	0.953 [0.93, 0.97]	0.972	0.957	0.956 [0.93, 0.98]	0.964	0.959	0.990
Transformer	FD003	0.870	0.900 [0.85, 0.95]	0.916	0.930	0.908 [0.87, 0.95]	0.928	0.936	0.939
Transformer	FD004	0.911	0.920 [0.89, 0.95]	0.937	0.932	0.932 [0.90, 0.96]	0.944	0.945	0.979

S10. Decision sensitivity, signature alignment, channel subsets and single channels (SD scale)

Bound level, tolerance band and lead time. Table 28 repeats the SD-scale analysis for $\alpha \in \{0.10, 0.05, 0.01\}$, tolerance bands $\tau \in \{10, 20, 40\}$ cycles and, within each cell, $L \in \{10, 20, 30\}$. Direction columns count backbone–sub-dataset–lead-time–mechanism combinations (8 cells $\times$ 3 lead times $\times$ 5 mechanisms). Ranking and regret columns count the 96 groups (8 cells $\times$ 3 lead times $\times$ 4 conditions: clean, $+1$ SD ramp, ± 2 SD offset, three-channel $+2$ SD ramp) in which the five mechanisms are ranked by $c_{\text{late}} r_{\text{late}} + r_{\text{early}}$ for $c_{\text{late}} \in \{5, 20, 100\}$; regret is the extra cost of choosing the mechanism that is best under one cost ratio when another is true, relative to the best cost. The full grid is in `sd_sensitivity_grid.csv`.

Table 28. Stability of the direction pattern and of the cost ranking across bound level and tolerance band (heteroscedastic head for the premature range; all mechanisms for the counts).

α	Band	$+2$ SD: premature $>$ unrec.	-2 SD: unrec. $>$ premature	Premature, $+2$ SD (%)	Ranking changes	Top changes / regret $> 10\%$
0.10	10	120/120	120/120	34–79	51/96	37 / 30
0.10	20	120/120	120/120	31–75	48/96	32 / 25
0.10	40	120/120	120/120	21–62	40/96	28 / 23
0.05	10	120/120	120/120	40–79	41/96	26 / 21
0.05	20	120/120	120/120	37–75	41/96	26 / 21
0.05	40	120/120	120/120	26–62	40/96	22 / 16
0.01	10	120/120	120/120	51–79	33/96	27 / 26
0.01	20	120/120	120/120	48–75	33/96	24 / 23
0.01	40	120/120	120/120	38–62	35/96	23 / 22

Scenarios. Aligned: every channel ramped by $+k$ SD in the direction of its own degradation trend (sign of the Spearman correlation between condition-centred reading and $-$RUL on the fitting engines). Anti: the opposite signs. Offset: constant $\pm k$ SD with random sign per channel (five realisations). 1 ch./3 ch.: $+2$ SD ramp on one or three randomly chosen sensor channels (ten draws). Rate: drift of 0.02 SD per recorded cycle accumulated from the first cycle of each test trajectory, all channels positive; Rate al.: the same with degradation-aligned signs.

Table 29. Heteroscedastic head, $L = 20$: premature / unrecognised (% of evaluated engines) by scenario; numbers after the scenario name are SD.

Cell	Aligned 1	Aligned 2	Anti 1	Anti 2	Offset 1	Offset 2	1 ch.	3 ch.	Rate	Rate al.
LSTM FD001	61 / 0.0	72 / 0.0	0 / 19.0	0 / 19.0	2 / 1.2	8 / 5.2	1 / 1.2	8 / 0.7	37 / 0.0	61 / 0.0
LSTM FD002	56 / 0.0	70 / 0.0	0 / 13.9	0 / 21.3	4 / 0.4	21 / 3.9	8 / 0.1	24 / 0.0	58 / 0.0	60 / 0.0
LSTM FD003	65 / 0.0	75 / 0.0	0 / 10.8	0 / 13.4	4 / 1.2	17 / 2.0	3 / 0.4	21 / 0.1	52 / 0.0	68 / 0.0
LSTM FD004	62 / 0.0	70 / 0.0	0 / 12.1	0 / 16.7	10 / 1.6	29 / 2.3	15 / 0.1	31 / 0.0	61 / 0.0	61 / 0.0
Transformer FD001	64 / 0.0	72 / 0.0	0 / 19.0	0 / 19.0	0 / 0.2	6 / 5.5	1 / 1.0	9 / 0.9	40 / 0.0	61 / 0.0
Transformer FD002	43 / 0.0	69 / 0.0	0 / 12.2	0 / 18.7	4 / 0.9	22 / 5.1	7 / 0.2	22 / 0.0	58 / 0.0	58 / 0.0
Transformer FD003	74 / 0.0	75 / 0.0	0 / 13.0	0 / 14.0	3 / 1.1	19 / 0.2	3 / 0.0	25 / 0.0	65 / 0.0	73 / 0.0
Transformer FD004	57 / 0.0	70 / 0.0	0 / 8.0	0 / 16.5	8 / 2.0	25 / 2.0	10 / 0.0	29 / 0.0	60 / 0.0	60 / 0.0

Table 30. Channel subsets: premature triggers (%, heteroscedastic head, $L = 20$), median [minimum, maximum] over ten random draws of the perturbed channels.

Cell	One channel, +2 SD	Three channels, +2 SD
LSTM FD001	0.0 [0.0, 3.8]	3.9 [0.0, 24.0]
LSTM FD002	6.7 [0.7, 27.5]	19.6 [2.7, 54.2]
LSTM FD003	3.4 [0.2, 8.6]	20.5 [0.4, 37.8]
LSTM FD004	3.9 [1.7, 52.2]	37.7 [2.1, 53.6]
Transformer FD001	0.0 [0.0, 1.8]	3.2 [0.0, 33.6]
Transformer FD002	3.3 [0.0, 22.4]	19.3 [1.2, 53.4]
Transformer FD003	1.4 [0.2, 10.0]	24.6 [1.4, 46.6]
Transformer FD004	4.4 [2.1, 33.4]	26.8 [2.7, 63.5]

Table 31. All five mechanisms under degradation-aligned and anti-aligned 2 SD ramps, $L = 20$ (% of evaluated engines). The aligned ramp drives every mechanism to the ceiling set by the share of engines with $y > L + \tau$.

Cell	Fixed var.	Heterosc.	MC Drop.	Ensemble	CP-norm
2 SD aligned, premature (%)					
LSTM FD001	72.0	72.0	72.0	72.0	72.0
LSTM FD002	70.0	70.0	70.0	70.0	70.0
LSTM FD003	75.0	75.0	75.0	75.0	75.0
LSTM FD004	69.6	69.6	69.6	69.6	69.6
Transformer FD001	72.0	72.0	72.0	72.0	72.0
Transformer FD002	70.0	69.5	70.0	70.0	69.6
Transformer FD003	75.0	75.0	75.0	75.0	75.0
Transformer FD004	69.0	69.6	69.3	69.6	69.5
2 SD anti-aligned, unrecognised (%)					
LSTM FD001	19.0	19.0	19.0	19.0	19.0
LSTM FD002	21.3	21.3	21.3	21.3	21.2
LSTM FD003	11.4	13.4	11.0	12.0	13.4
LSTM FD004	16.9	16.7	16.9	16.9	16.6
Transformer FD001	19.0	19.0	19.0	19.0	19.0
Transformer FD002	20.0	18.7	20.2	15.0	18.3
Transformer FD003	13.8	14.0	13.8	14.0	14.0
Transformer FD004	15.4	16.5	16.1	16.5	16.5

Table 32. Single-channel ±2 SD ramps on the final window, heteroscedastic head, $L = 20$: premature / unrecognised (%). Columns: sub-dataset digit and backbone (L LSTM, T Transformer); ·: channel constant on that sub-dataset and not used.

Channel	Sign	1L	1T	2L	2T	3L	3T	4L	4T
T24	+	1/0	1/0	2/0	0/0	1/0	1/0	2/0	2/0
	−	0/3	0/2	6/0	3/0	0/1	0/0	6/0	5/0
T30	+	1/0	0/0	9/0	5/0	3/0	0/0	10/0	6/0
	−	0/1	0/0	1/0	0/0	0/0	1/0	2/0	1/0
T50	+	1/0	1/0	28/0	22/0	4/0	5/0	36/0	18/0
	−	0/4	0/3	0/3	0/4	0/1	0/1	1/5	1/2
P15	+	·	·	4/0	2/0	·	·	4/0	3/0
	−	·	·	4/0	2/0	·	·	4/0	3/0
P30	+	0/2	0/2	4/0	2/0	4/0	5/0	4/0	3/0
	−	1/0	1/0	4/0	1/0	1/1	2/1	4/0	3/0
Nf	+	4/0	2/0	4/0	1/0	6/0	10/0	4/0	3/0
	−	0/3	0/3	4/0	2/0	0/2	0/2	4/0	3/0
Nc	+	2/0	1/0	1/1	0/2	0/0	1/0	2/0	2/0
	−	0/2	0/1	12/0	15/0	0/1	1/0	9/0	8/0
Ps30	+	5/0	6/0	44/0	41/0	7/0	11/0	52/0	33/0
	−	0/6	0/8	0/7	0/9	0/2	0/2	0/9	0/7
phi	+	0/4	0/4	4/0	2/0	9/0	10/0	4/0	3/0
	−	3/0	3/0	4/0	2/0	0/3	0/2	3/0	3/0
NRf	+	2/0	0/0	4/0	2/0	9/0	12/0	4/0	3/0
	−	0/2	0/2	4/0	2/0	0/2	0/2	4/0	3/0
NRc	+	1/0	0/0	11/0	13/0	1/0	0/0	41/0	61/0
	−	0/1	0/1	6/1	11/1	0/1	1/0	1/2	14/3
BPR	+	4/0	1/0	21/0	17/0	7/0	11/0	2/0	12/0
	−	0/3	0/2	0/3	0/4	1/1	0/1	9/0	5/1
htBleed	+	0/0	0/0	11/0	7/0	4/0	4/0	13/0	7/0
	−	0/0	0/0	1/0	0/0	0/2	0/0	1/1	2/0
W31	+	0/0	0/1	3/0	1/0	0/2	1/0	4/0	3/0
	−	0/0	0/0	5/0	2/0	2/0	3/0	4/0	3/0
W32	+	0/2	0/1	3/0	1/0	0/0	0/0	4/0	3/0
	−	1/0	0/0	5/0	2/0	4/0	7/0	4/0	3/0

S11. First trigger along test trajectories

Each test trajectory is evaluated at every window, and the engine is scored by its first trigger ($b \le L$). *Premature first*: the first trigger falls at true RUL $> L + \tau$; *timely first*: at $y \le L + \tau$; *never*: no trigger before the record ends. Engines reaching L: test trajectories whose last label is $\le L$, the only ones on which a missed need can be observed; *missed among reaching*: share of those with no trigger by the time $y \le L$. Forgone: mean RUL at a premature first trigger. Drift accumulates at 0.02 SD per cycle from the first recorded cycle (all channels positive, degradation-aligned or anti-aligned); heteroscedastic head, $L = 20$, $\tau = 20$; rates are means over the five models.

Cell	Scenario	Premature first (%)	Timely first (%)	Never (%)	Reaching L	Missed among reaching (%)	Forgone (cycles)
LSTM FD001	clean	1.6	23.2	75.2	19	1	23
	+ all	65.2	1.2	33.6	19	0	93
	aligned	91.0	0.0	9.0	19	0	128
	anti-aligned	0.0	0.0	100.0	19	100	–
LSTM FD002	clean	21.7	13.1	65.1	54	0	47
	+ all	90.3	0.0	9.7	54	0	139
	aligned	91.2	0.0	8.8	54	0	141
	anti-aligned	3.6	1.3	95.0	54	87	81
LSTM FD003	clean	0.4	19.2	80.4	14	0	40
	+ all	79.2	0.0	20.8	14	0	137
	aligned	93.4	0.0	6.6	14	0	158
	anti-aligned	0.0	0.4	99.6	14	97	–
LSTM FD004	clean	26.4	10.0	63.5	40	0	59
	+ all	94.7	0.0	5.3	40	0	181
	aligned	94.9	0.0	5.1	40	0	182
	anti-aligned	0.1	0.1	99.8	40	100	201
Transformer FD001	clean	2.2	23.2	74.6	19	0	24
	+ all	68.8	0.4	30.8	19	0	95
	aligned	91.6	0.0	8.4	19	0	130
	anti-aligned	0.0	0.0	100.0	19	100	–
Transformer FD002	clean	16.8	15.2	68.0	54	0	44
	+ all	89.6	0.0	10.4	54	0	136
	aligned	90.3	0.0	9.7	54	0	137
	anti-aligned	0.0	0.7	99.3	54	97	–
Transformer FD003	clean	1.0	21.2	77.8	14	0	37
	+ all	92.8	0.0	7.2	14	0	159
	aligned	98.8	0.0	1.2	14	0	176
	anti-aligned	0.0	0.0	100.0	14	100	–
Transformer FD004	clean	24.1	11.0	64.9	40	0	52
	+ all	92.9	0.0	7.1	40	0	174
	aligned	92.7	0.0	7.3	40	0	174
	anti-aligned	0.3	0.8	98.8	40	95	186